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Decarbonisation in Europe: Modelling Economic Feasibility and the Glidepath for Gas

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Executive Summary

Modelling multiple decarbonisation pathways for Europe to 2050 illuminates key trade-offs between emissions reduction, costs and the implications for gas demand over the period. The most ambitious scenario that front-loads decarbonisation to 2040 and achieves net zero by 2050 indicates prohibitively high marginal abatement costs, while a linear pathway to net zero reduces those costs sharply, achieving the second deepest reductions in gas demand over the period. Scenarios that do not achieve net zero emissions by 2050 reduce those costs further and keep a larger role for gas in the energy mix for longer.

The critical political context is the European Commission's proposal for a 90% reduction in net greenhouse gas (GHG) emissions by 2040 compared to 1990, positioning it as an intermediate milestone towards achieving net zero by 2050. Initially proposed in February 2024, the formal legislative proposal to enshrine this target into EU law has been delayed, with publication expected in Spring 2025. However, achieving such deep decarbonisation comes at a significant cost, particularly in the absence of breakthrough technologies that require substantial public and private financial backing to be developed and deployed at scale and acceptable cost.

In the "Accelerated Path to Net Zero" scenario modelled here that broadly tracks the EC proposal, reducing net GHG emissions by at least 90% by 2040 results in prohibitively high marginal abatement costs (MAC), reaching **€17,246/tCO_2e** by 2040. In contrast, a more gradual "Linear Path to Net Zero" (55% reduction in 2030, 76% in 2040, and net zero in 2050) spreads costs more efficiently over time, with MAC at €420/tCO_2e in 2040 and **€1,944/tCO_2e** in 2050. Even this MAC of €1,944/tCO_2e remains very high, highlighting the substantial cost required to eliminate the last remaining emissions as low-cost mitigation options become exhausted.

However, it is crucial to recognise that the alternative – inaction or insufficient mitigation – also carries substantial societal costs[1] associated with continued emissions, even though these costs are challenging to quantify precisely in practice. Our projections indicate significant uncertainty in societal cost estimates for 2050, ranging from approximately €236/tCO_2 to €3,938/tCO_2. Consequently, direct comparisons with marginal abatement costs become difficult; for instance, even the relatively lower MAC of €1,944/tCO_2e in the "Linear Path to Net Zero" scenario aligns with societal costs only under particular discount rates and growth assumptions.

Achieving significant decarbonisation at a lower cost is possible in scenarios that do not require net zero by 2050. The Baseline scenario achieves an 84% reduction in emissions by 2050 with a significantly lower carbon price of €318/tCO_2e, while the High Carbon Price scenario attains an 86% reduction at €445/tCO_2e. These figures highlight that a significant reduction in emissions can be accomplished at a fraction of the cost of full net zero, suggesting that pursuing net zero through purely domestic abatement measures leads to exponentially rising costs. Thus, integrating and linking international carbon markets and low-carbon energy imports, such as hydrogen, bioenergy, and synthetic fuels, could provide a more cost-effective alternative. A diversified decarbonisation strategy that leverages external low-carbon resources can mitigate the economic impact of stringent domestic emissions reductions.

[1] A meausre of such costs in the climate economics literature is the Social Cost of Carbon (SCC). SCC provides a measure of these damages, though estimates vary significantly based on assumptions about economic growth, climate sensitivity, and discount rates.

	GHG emissions, $mtCO_{2e}$ (% reduction relative to 1990)			Marginal Abatement (Carbon) Cost*, €/tCO_{2e}		
	2030	2040	2050	2030	2040	2050
Baseline	2,793 (52%)	1,935 (67%)	908 (84%)	68	193	318
High Carbon Price	2,793 (52%)	1,767 (70%)	833 (86%)	68	257	445
Low Carbon Price	2,793 (52%)	2,285 (61%)	1,878 (68%)	68	91	114
Accelerated Path to Net Zero	2,622 (55%)	583 (90%)	0 (100%)	134	17,246	426
Linear Path to Net Zero	2,622 (55%)	1,407 (76%)	0 (100%)	134	420	1,944

Notes: Model **inputs** are coloured in **black**, while outputs are in red font; * The marginal abatement cost (MAC) in the first three scenarios (Baseline, High Carbon Price, and Low Carbon Price) reflects the assumed carbon price, which can approximate the MAC if the market functions efficiently. In contrast, the last two scenarios (Accelerated Path to Net Zero and Linear Path to Net Zero) impose a binding GHG budget, with the MAC derived from the shadow price of this constraint, representing the least-cost pathway to achieving the specified emissions reductions

In the five scenarios modelled here, the demand for gas in Europe (EU27, UK, Norway, and Switzerland) declines differently, responding to policies and market dynamics. In particular, in 2050, gas demand varies widely:

- *Baseline scenario*: 291 bcm (vs 407 bcm in 2023), reflecting a continued role for gas in non-net-zero decarbonisation (84% reduction in GHG emissions relative to 1990).
- *High Carbon Price scenario*: 250 bcm, with a slightly more aggressive reduction (86% reduction in GHG emissions relative to 1990) driven by high carbon costs but without politically mandated targets.
- *Low Carbon Price scenario*: 370 bcm, indicating a slower transition (68% reduction in GHG emissions relative to 1990) due to weaker carbon pricing incentives.
- *Linear Path to Net Zero*: 160 bcm, illustrating a decline in gas use driven by the political ambition of reaching net zero gradually from 2030.
- *Accelerated Path to Net Zero*: 141 bcm, showing the impact of the aspirational goal of front-loading reductions to an early period (90% by 2040) with the potential to lock into inferior abatement solutions.

Gas use in power generation falls post-2040, particularly in net-zero scenarios, as renewables with strong public support take over. However, gas remains a key feedstock for hydrogen production and industry across all pathways. In the Baseline and High Carbon Price scenarios, 50–60% of transformation gas is allocated to hydrogen production with CCS. In the Low Carbon Price scenario, gas remains dominant across multiple sectors, with lower low-carbon hydrogen uptake due to weaker carbon price incentives. In the Linear and Accelerated Path to Net Zero scenarios, gas use for hydrogen production eventually declines, reflecting the shift toward green hydrogen produced via electrolysis due to residual emissions from low-carbon hydrogen incompatible with policy targets of net zero emissions.

Final consumption sectors, including industry, buildings, and transport, experience a more gradual decline in gas usage due to high operational and capital investment costs of alternatives, especially in the medium term (by 2040). In all scenarios, switching costs in these sectors are relatively high without additional policy support. The Low Carbon Price scenario exhibits the slowest transition, with gas remaining prevalent in industrial and residential heating applications even by 2050.

Gas demand elasticity relative to carbon prices remains low, particularly before 2040. Even under high carbon prices of €445/tCO_2e in 2050, gas remains in power and industrial use, indicating that carbon pricing alone cannot drive progress toward net zero. Thus, complementary policies such as financial support for low-carbon energies and infrastructure and environmental performance standards are necessary to accelerate the transition, particularly in final consumption sectors where infrastructure lock-in and high switching costs remain significant barriers.

The modelling suggests that while gas demand declines across all scenarios, the extent and speed of reduction depend on policy stringency, public and financial support, availability of carbon capture, utilisation and storage (CCUS), and the pace of renewable and low-carbon hydrogen adoption. In deep decarbonisation pathways, gas transitions from a direct combustion fuel to a feedstock for hydrogen production, particularly in pathways where CCUS is feasible at lower costs.

The results and analysis suggest the following policy recommendations:

1. *A phased approach to the energy transition* minimises economic shocks, avoiding extreme price spikes and locking into inferior abatement measures. A gradual emissions reduction pathway reduces near-term costs and allows time for technological advancements.
2. *Leverage international carbon markets and low-carbon energy imports.* Incorporating external abatement solutions, such as linking to other international emissions trading and importing renewable, low-carbon hydrogen and bioenergy, can reduce the cost burden of domestic decarbonisation.
3. *Develop hydrogen infrastructure* as a bridge for reducing dependence on high emissions technologies. A well-structured transition to renewable and low-carbon hydrogen enables deeper decarbonisation while avoiding abrupt disruptions to energy supply.
4. *Support innovation in renewables, CCUS, and synthetic fuels*. Targeted R&D and financial incentives can mitigate abatement cost escalation and enhance technological readiness for large-scale deployment.
5. *Complement carbon pricing with sector-specific policies*. Electrification incentives, demand-side flexibility, and direct industrial decarbonisation measures are required to accelerate the transition.

In short, significant decarbonisation is achievable in Europe at a moderate cost. However, net zero requires substantial public and financial support and, crucially, technological advancements at a pace consistent with aspirational political goals. International abatement solutions and low-carbon imported fuels could provide a cost-effective alternative to achieving net zero climate goals without imposing excessive economic burdens. Policymakers must balance ambition and economic feasibility, ensuring long-term decarbonisation strategies reflect overall economic development conditions, the underlying costs of meeting those targets, and the technological pace.

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1. Introduction

Energy system optimisation models have been crucial in helping to explore energy strategies and scenarios toward efficient and cost-effective deep decarbonisation[2]. However, there is an increasing need to incorporate new features that explicitly capture the critical role of flexibility in managing demand and supply variations[3] and technology and geopolitical risks in the ongoing decarbonisation efforts.

The energy crisis of 2021–2023 exposed several critical vulnerabilities in the European energy system, particularly the risks associated with energy supply and demand imbalances. Low levels of renewable energy generation – stemming from reduced wind speeds, prolonged droughts affecting hydropower, and a downturn in nuclear power production due to maintenance and unforeseen outages – coincided with the interruption of natural gas supplies from Russia. This convergence of factors severely strained Europe's energy security, prompting a reliance on more carbon-intensive sources (e.g., coal) and unprecedented volatility in energy prices. The crisis points to the inherent risks of an over-dependence on individual energy sources and the necessity of diversifying the supply mix for managing supply and demand variability. Furthermore, it highlighted the urgent need for flexibility within the energy system to mitigate the impacts of both variability in renewable energy production and geopolitical shocks.

To assess the economic implications and trade-offs associated with these challenges, this paper introduces the Energy Futures Optimisation Model (EFOM), a detailed simulation tool designed to analyse the evolution of energy systems, with a particular emphasis on the role of policy instruments, flexibility sources and energy supply security. EFOM represents an evolution of the previous version of the energy system optimisation model[4]. EFOM contributes to the energy system modelling literature by offering an analytically rigorous framework to examine the trade-offs between achieving climate targets, ensuring energy security, and minimising costs under diverse geopolitical and technological risks. It does so by explicitly modelling various flexibility mechanisms – such as energy storage, international energy trade, demand-side response, and cross-sectoral flexibility – and key policy instruments, such as performance and technical standards, carbon tax and budgets, and supply subsidies.

EFOM has here been applied to the European energy system[5] to demonstrate the model's capabilities, simulating five distinct scenarios reflecting variability in key decarbonisation policy instruments, such as carbon pricing, RES subsidies and policy-driven climate targets (Table 1). These scenarios were designed to assess the impact of policy instruments on decarbonising the European energy system, evaluate the costs associated with policy targets, and examine the role of natural gas, low-carbon technologies, and renewable energy sources.

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[2] see, e.g., Capros et al. (2019) on energy system modelling of the EU's carbon neutrality strategy; Bistline (2021a) for a summary of insights from national net-zero energy system modelling studies; Browning et al. (2023) for comparison of net zero emissions scenario modelling efforts, part of the EMF37, for North America; Sarmiento et al. (2024) for a transatlantic net zero pathways comparison between EMF and ECEM models; Boitier et al. (2023) and Henke et al. (2024) for an inter-model comparison of the EU's net zero pathways modelling; Vats and Mathur (2022) on the application of the MARKAL model for the analysis of net zero pathway for India; Fragkos et al. (2021) for energy system transition pathways modelling for Australia, Brazil, China, India, Indonesia, Japan, Republic of Korea, Russia and other regions

[3] see, e.g., Bistline's (2021a; b) discussion on energy system modelling research needs to tackle and manage increasing variability of renewable energy supply, RES, and energy demand

[4] The first version of the model was published by Chyong et al. (2024)

[5] In fact, the model if flexible and can be applied to other countries and regions.

Table 1: Scenarios modelled

Scenario Name	GHG Constraint	Carbon Price	RES Subsidy	Key Features
Baseline	No	Moderate	Yes	Reference scenario, reflecting an existing approach to decarbonise
High Carbon Price	No	High	Yes	Rising carbon price consistent with net zero modelling by the European Commission
Low Carbon Price	No	Low	Yes	Sensitivity analysis with a low carbon price
Accelerated Path to Net Zero	55% by 2030, 90% by 2040 and net zero by 2050	No	No	Accelerated policy-driven GHG reduction pathway, no carbon price or subsidies
Linear Path to Net Zero	55% by 2030, 76% by 2040, and net zero by 2050	No	No	Linear GHG reduction towards net zero, no carbon price or subsidies

Notes: "Yes" – the policy variable included in the scenario for modelling; otherwise, "No"

The **Baseline scenario** serves as a reference, projecting the evolution of the European energy system along a decarbonisation pathway that uses only carbon price and renewable subsidies as key instruments to decarbonise the energy system. It includes existing subsidies for renewable energy and a moderate carbon price trajectory (see Appendix 3 for details**Error! Reference source not found.**) and no specific greenhouse g as (GHG) emissions targets for the whole economy. This scenario aims to provide a benchmark for understanding the impact of alternative policy measures. It assumes that renewable subsidies will be gradually phased out by 2050.

The **High Carbon Price scenario** assumes a carbon price that rises to a level consistent with achieving deeper emissions reduction by 2050. As in the Baseline, this scenario includes renewable energy subsidies, which are assumed to be phased out gradually by 2050. The primary focus is understanding how a rising carbon price trajectory influences the decarbonisation pathway, particularly in fuel switching and technology deployment.

The **Low Carbon Price scenario** explores the impact of a relatively low carbon price trajectory on the decarbonisation pathway. Similar to the previous two scenarios, it includes the gradual phase-out of renewable energy subsidies by 2050. This scenario assesses the challenges and potential limitations of significantly reducing GHG emissions under a less ambitious climate policy.

The **Accelerated Path to Net Zero Scenario** sets stringent economy-wide GHG reduction targets, aiming for a 55% reduction by 2030, a 90% reduction by 2040 and reaching net zero by 2050. This scenario considers these three policy targets as given, implying a policy-driven accelerated trajectory to reach net zero by 2050. Unlike the Baseline scenario, it does not include renewable energy subsidies or a carbon price. Considering the ambitious emissions constraints, this scenario aims to "back out" an economy-wide marginal carbon and energy costs trajectory to reach these policy objectives.

The **Linear Path to Net Zero Scenario** is identical to the Accelerated Path scenario with the exception that the GHG reduction target is linearly extrapolated from the 2030 target (55% reduction) to reach net zero by 2050, resulting in a 76% reduction by 2040 instead of a policy target reducing GHG emissions by 90%. This scenario aims to determine an economy-wide marginal carbon and energy costs required to reach net zero under a linear GHG reduction trend, contrasting the much more rapid and stringent policy-driven approach of the Accelerated Path scenario. This scenario does not include renewable energy subsidies or a carbon price.

The rest of this paper is structured as follows. It discusses key insights from the modelling in the following two sections. In the last section, it offers concluding remarks, outlining key findings, limitations and future work required. The various appendices discuss the modelling framework (Appendix 1), data inputs and scenarios (Appendix 2 and 3), and detailed model calibration results for 2030 and 2050 (Appendix 4).

2. Cost of meeting the climate targets in policy-driven scenarios

Two policy-driven scenarios are modelled: (i) the first one (accelerated path to net zero) aims to meet 55% GHG reduction in 2030, 90% by 2040 and net zero by 2050, while (ii) the second aims to meet 55% reduction by 2030 and net zero by 2050 while 2040 emissions target is implied from linear interpolation between 2030 and 2050 targets (hence the name – linear path to net zero). A critical output from these two policy-driven scenarios is the estimated shadow price of the GHG emissions constraint, or, in other words, the marginal abatement (carbon) cost of achieving incremental GHG reduction. In contrast to the three carbon pricing scenarios – Baseline, High and Low Carbon prices – that assume marginal carbon cost as inputs to drive the energy system decarbonisation, these two-policy-driven scenarios constrain the model to achieve those targets by a specific time, given technology roll-out constraints, their cost evolution (for details on technology availability and costs, see Appendix 2) and demand projections (for energy demand projection from the scenarios, see Appendix A.2.2. and Table 4).

The GHG reduction trends and marginal and average abatement (carbon) costs across the five scenarios are reported in Table 2.

Table 2: GHG Emissions and marginal carbon cost

	GHG emissions, $mtCO_{2e}$ (% reduction relative to 1990)			Marginal Abatement (Carbon) Cost***, €/tCO_{2e}			Average*Abatement Cost, €/tCO_{2e}		
	2030	2040	2050	2030	2040	2050	2030	2040	2050
Baseline**	2,793 (52%)	1,935 (67%)	908 (84%)	68	193	318	n.a.	69	116
High Carbon Price**	2,793 (52%)	1,767 (70%)	833 (86%)	68	257	445	n.a.	125	145
Low Carbon Price**	2,793 (52%)	2,285 (61%)	1,878 (68%)	68	91	114	n.a.	n.a.	n.a.
Accelerated Path to Net Zero	2,622 (55%)	583 (90%)	0 (100%)	134	17,246	426	71	2,499	872
Linear Path to Net Zero	2,622 (55%)	1,407 (76%)	0 (100%)	134	420	1,944	71	65	271

Notes: Model **inputs** are coloured in **black**, while outputs are in red font; * average cost (capex and opex) is computed relative to the low carbon price scenario; ** Carbon price assumed in these three scenarios shown here are for ETS1 (for further information, see Appendix 3); *** The marginal abatement cost (MAC) in the first three scenarios (Baseline, High Carbon Price, and Low Carbon Price) reflects the assumed carbon price, which can approximate the MAC if the market functions efficiently. In contrast, the last two scenarios (Accelerated Path to Net Zero and Linear Path to Net Zero) impose a binding GHG budget, with the MAC derived from the shadow price of this constraint, representing the least-cost pathway to achieving the specified emissions reductions

The marginal abatement cost (MAC[6]) represents the incremental cost of reducing one additional unit of CO_2 emissions at a given time, providing a price signal for incremental abatement decisions. In contrast, the average abatement cost (AAC) reflects the total cost per unit of emissions reduced, averaged over all abatement efforts in a given scenario. The AAC is computed relative to the Low Carbon Price scenario, which represents the least stringent policy pathway modelled and provides a benchmark for assessing the cost-effectiveness of alternatively more ambitious mitigation scenarios.

[6] This refers to the incremental cost of reducing an additional (marginal) unit of emissions. It represents the financial incentive for emitters to undertake additional abatement and is directly tied to the carbon pricing policies such as carbon taxing or pricing carbon emissions via a market-based GHG cap and permit trading system like the EU ETS.

Thus, MAC helps policymakers and investors assess the economic feasibility of further reductions and informs policy design, while AAC reflects the total cost per unit of emissions reduced. AAC measures cost-effectiveness, showing whether a decarbonisation pathway minimises overall costs or leads to excessive expenditure over the transition period.

2.1 2030 target

The Baseline, High Carbon Price, and Low Carbon Price scenarios achieve a 52% reduction in GHG emissions from 1990 levels, with total emissions of 2,793 MtCO_2e in 2030. These scenarios use assumed marginal carbon costs (a proxy for the EU ETS prices, see Appendix 3) as inputs rather than imposing direct emissions caps. By contrast, the Accelerated Path and Linear Path to Net Zero scenarios apply a policy-driven GHG emissions cap, achieving a 55% reduction (2,622 MtCO_2e) by 2030.

A critical insight based on the assumptions and the model is that this 3-percentage-point additional reduction requires a near doubling of the marginal carbon cost, increasing from €68/tCO_2e (carbon price scenarios) to €134/tCO_2e (net zero scenarios) in 2030 (see Table 2). To put this finding in the context of the EU ETS carbon market, in 2024, the average EU ETS carbon price was €66/tCO_2 in 2022/23 - €83/tCO_2. The sharp cost increase highlights the rising cost of incremental decarbonisation, as deeper reductions in the near term require more aggressive interventions within a short timeframe.

While MAC is crucial as it indicates the incremental cost of reducing the next unit of emissions, the AAC measures the average cost (capex and opex) of all incremental actions taken to reduce that 3-percentage-point in 2030 with the assumed cost trends (see Chyong et al., 2024 and Appendix 2); thus, to move from 52% to 55% emissions reduction in 2030 requires an average of €71/tCO_2e across both net zero scenarios as both start from the same set of assumptions for 2030. The divergence in abatement costs, however, starts from 2040.

2.2 2040 target

By 2040, the Baseline scenario achieves a 67% reduction (1,935 MtCO_2e) at an assumed carbon price (ETS1) of €193/tCO_2e, while the High Carbon Price scenario improves slightly to a 70% reduction (1,767 MtCO_2e) at €257/tCO_2e. The Low Carbon Price scenario lags at a 61% reduction (2,285 MtCO_2e) with a much lower assumed carbon price of €91/tCO_2e, indicating that weaker pricing results in slower decarbonisation.

In contrast, the Linear Path to Net Zero scenario achieves a 76% reduction (1,407 MtCO_2e) at a marginal carbon cost of €420/tCO_2e. By design, the Accelerated Path to Net Zero scenario drives emissions down to 583 MtCO_2e (90% reduction) but at an extremely high marginal carbon cost of €17,246/tCO_2e. The dramatic spike in marginal cost in the Accelerated Path scenario reflects the difficulty of rapid, deep decarbonisation, requiring costly abatement measures and extensive structural changes within the energy system modelled here. The Linear Path scenario follows a more gradual trajectory, maintaining lower costs than the Accelerated Path but still significantly higher than carbon price-driven pathways.

An interesting observation from the modelling is that in the Linear Path scenario, the AAC (€65/tCO_2e) is the lowest while achieving the highest decarbonisation, suggesting that a gradual transition spreads costs efficiently over time. In contrast, the Accelerated Path scenario sees an extreme AAC of €2,499/tCO_2e, indicating the high cumulative cost of rapid decarbonisation. Despite both net zero scenarios aiming for full decarbonisation by 2050, the Accelerated Path accumulates far greater costs by 2040, whereas the Linear Path maintains cost efficiency while still achieving similar reductions. It also suggests that the optimal solution to decarbonise based on the Linear Path achieves a lower total abatement cost than either of the Baseline and High carbon price scenarios, as those may contain a set of policies that drive the system to a less optimal configuration (e.g., capping ETS2 price at €54/tCO_2e, in line with the EC's current position with respect to the ETS2 market – see Appendix 3).

2.3 2050 target

By design, the Linear and Accelerated Path scenarios achieve net zero emissions by 2050, while the Baseline and High Carbon Price scenarios still retain 908 MtCO_2e (84% reduction relative to 1990) and 833 MtCO_2e (86% reduction relative to 1990) of residual emissions, respectively. The Low Carbon Price scenario lags further behind at 1,878 MtCO_2e (68% reduction relative to 1990), reinforcing that carbon pricing alone cannot drive deep decarbonisation.

The AAC values for 2050 outline the cost-effectiveness of the various pathways. The average abatement cost of moving from the 2050 emissions levels in the low carbon price scenario to reach net zero in line with the Linear Path scenario (€271/tCO_2e) remains significantly lower than the Accelerated Path (€872/tCO_2e), confirming that a gradual transition reduces cumulative costs. Despite achieving net zero in the same year, the Accelerated Path scenario's total abatement cost remains the highest, reflecting the long-term consequences of extreme front-loading. In contrast, the Baseline (€116/tCO_2e) and High Carbon Price (€145/tCO_2e) scenarios remain relatively cost-efficient but fall short of achieving full decarbonisation, illustrating that carbon pricing alone does not guarantee deep emissions cuts.

The marginal abatement cost trends in 2050 reinforce these findings. The Linear Path scenario sees its MAC rise to a staggering €1,944/tCO_2e, demonstrating the high cost of eliminating the final emissions. In contrast, the Accelerated Path scenario benefits from its early investments, bringing MAC down to €426/tCO_2e. However, the persistently high AAC in the Accelerated Path scenario suggests that its cost burden remained excessive despite lowering its final-stage MAC. This finding reflects what was already observed in 2040: an extreme MAC spike in the Accelerated Path scenario led to high cumulative (and sunk from the 2050 perspective) costs, whereas the Linear Path, despite its higher MAC in 2050, spread costs more efficiently over time. The Baseline and High Carbon Price scenarios, with MAC values of €318/tCO_2e and €445/tCO_2e, respectively, remain moderate in cost but leave residual emissions, confirming that pricing mechanisms alone, even at higher levels, do not drive deep emissions cuts. These results underscore the importance of balancing political ambition with economic cost efficiency. The proposed 90% GHG reduction target by 2040, while lowering future MAC, can come at a significant cumulative abatement cost if technological innovation is not supported at the pace aligned with the ambitious target.

However, it is crucial to recognise that the alternative – inaction or insufficient mitigation – also carries substantial, albeit difficult to quantify, societal costs associated with ongoing emissions. The Social Cost of Carbon (SCC) provides a measure of these damages, though estimates vary significantly based on assumptions about economic growth, climate sensitivity, and discount rates.[7] Comparing MAC with SCC allows us to assess whether the cost of reducing emissions aligns with the broader economic damages caused by those emissions. In simple terms, if MAC exceeds SCC, mitigation may be considered "too expensive" relative to the damages it avoids. If SCC exceeds MAC, mitigation is economically justified, as it costs less to reduce emissions than the damage they would otherwise cause.

Our projections based on Tol (2023) suggest that SCC values for 2050 range from approximately €236/tCO_2 (high PRTP[8] and low SCC growth[9]) to €3,938/tCO_2 (low PRTP and high SCC growth). These estimates illustrate SCC's wide uncertainty, making direct comparisons with MACs dependent on discounting and damage escalation assumptions. While SCC estimates at lower discount rates and higher growth assumptions justify strong mitigation policies, the MACs in the "Accelerated Path to Net Zero" scenario far exceed projected SCC values, particularly in 2040. Even in the more gradual "Linear Path to Net Zero," the 2050 MAC of €1,944/tCO_2e falls within the SCC range but only under specific discount rate and growth assumptions.

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[7] Recent literature highlights significant methodological uncertainties in SCC estimation, including variations in emissions trajectories, climate sensitivity, damage functions, and particularly the appropriate choice of discount rates, all contributing to substantial variability in SCC estimates (Weitzman, 2011; Pindyck, 2019; Dietz & Stern, 2015; Nordhaus, 2017; Ricke et al., 2018; Burke et al., 2015). Consequently, despite its theoretical appeal, the practical use of SCC remains contentious, limiting its reliability as a standalone guide for policy decisions.

[8] The Pure Rate of Time Preference (PRTP) reflects how much less we value future costs and benefits compared to today. A high PRTP (e.g., 3%) means that future climate damages are considered less important than present-day economic costs, leading to policies that prioritise short-term growth over long-term climate protection. In contrast, a low PRTP (e.g., 1% or 0%) assumes that future generations should be given equal consideration, making strong climate action today more justified. This choice matters because it affects how much weight we place on avoiding long-term climate damages when setting policies like carbon pricing and emission reduction targets.

[9] The SCC grows over time because the damages from each tonne of CO_2 increase as the climate warms. As economic activity expands, climate impacts – such as extreme weather, sea-level rise, and biodiversity loss – become more costly to society. Additionally, some climate effects may trigger irreversible tipping points, meaning future emissions could cause disproportionately higher damage.

3. Natural gas in decarbonisation pathways

The results from the five modelled scenarios show different energy demand trends, reflecting the impact of different policy and market conditions on the pace of decarbonisation. Across all scenarios, total energy demand declines progressively between 2030 and 2050 (Table 3), with variations in the reduction rate depending on the scenario and policies modelled.

The Baseline and High Carbon Price scenarios achieve a relatively high level of decarbonisation by 2050 (see Table 2). However, this requires a significant share of primary energy to be converted into low-carbon synthetic fuels (synfuels), leading to higher primary energy demand overall. Both scenarios result in similar overall energy demand trajectories, declining by approximately 31.3% and 31.2% below 2022 levels by 2050, respectively. By contrast, the Low Carbon Price scenario achieves a much lower level of decarbonisation by 2050, thereby reducing the need for large-scale transformation of primary energy into synfuels. The scenario initially follows a similar trajectory to the Baseline but exhibits a steeper decline post-2040, reaching 34.3% below 2022 levels by 2050. This finding suggests that lower synthetic fuel demand contributes to a lower primary energy requirement, highlighting the increasing role of hydrogen and its derivatives in deep decarbonisation pathways.

Similarly, the Accelerated Path to Net Zero scenario, which imposes a more ambitious and aspirational emissions reduction target (e.g., reduction of 90% by 2040) to drive deep decarbonisation, exhibits the sharpest decline in total energy demand, reaching 39% below 2022 levels by 2040, before bouncing back at a 29.3% reduction by 2050. Reliance on earlier synfuel investments will increase primary energy consumption (the lock-in effect) in this scenario by 2050. In contrast, the Linear Path to Net Zero scenario follows a more gradual reduction trajectory, reaching 28% below 2022 levels by 2040 and 30.4% by 2050. Although it achieves similar decarbonisation levels as the Accelerated Path scenario by 2050, the gradual decarbonisation approach implies lower medium-term (2030-40) demand for synfuels, leading to a slightly lower primary energy requirement in 2050 (avoiding the early synfuel lock-in effect).

Compared to the IEA scenarios, the modelled scenarios align more closely with the IEA APS trajectory regarding energy demand reduction, which reaches 30.6% below 2022 levels by 2050. Due to its stringent emissions constraints modelled, the Accelerated Path to Net Zero scenario, by design, exhibits a significantly steeper decline by 2040 than both IEA APS and the other modelled scenarios.

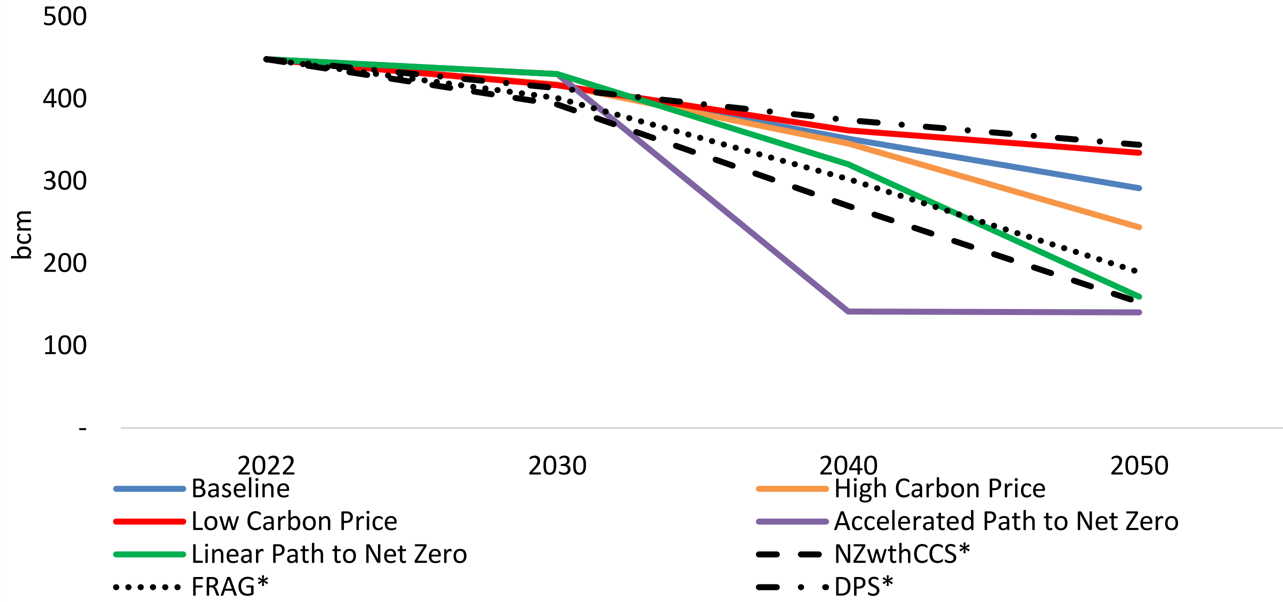
Table 3: Projected energy demand (PJ)

	2022	2030	2040	2050
Baseline	63121	48,452	44,363	43,364
High Carbon Price		48,464	45,154	43,410
Low Carbon Price		48,241	44,089	41,496
Accelerated Path to Net Zero		48,562	38,768	44,609
Linear Path to Net Zero		48,562	45,350	43,961
IEA STEPS	54704	51,700	45,817	43,145
IEA APS		49,535	41,330	37,975

Notes: IEA (2023) data is for the EU27, whereas our modelling results cover the EU27, UK, Norway and Switzerland

Regarding gas demand projections, the trends across the five modelled scenarios show varying policy impacts (Figure 1). The three carbon price scenarios (Baseline, High Carbon Price, and Low Carbon Price) exhibit moderate declines in gas demand relative to 2022, with reductions of -19.30% to -22.82% by 2040 and -25.38% to -45.56% by 2050. The High Carbon Price scenario leads to the steepest decline within this group, reducing gas consumption by -45.5% by 2050, whereas the Low Carbon Price scenario shows the slowest transition, with demand decreasing only -25.4%. The Baseline scenario follows an intermediate trajectory, reaching -35.0% by 2050. The higher carbon price accelerates gas displacement, particularly after 2040, but the differences between the Baseline and High Carbon Price scenarios remain modest until 2040.

Figure 1: Gas demand projections by scenarios



Notes: * OIES Energy Transition Scenarios 2024, for details, see Fulwood (2024) Energy Transition Scenarios: Impact on Natural Gas, available here: https://www.oxfordenergy.org/publications/energy-transition-scenarios-impact-on-natural-gas/

When comparing the Baseline scenario with the two net-zero scenarios, the divergence becomes more pronounced beyond 2040. While the Baseline scenario still retains a significant role for gas (291 bcm) in 2050, the Linear Path to Net Zero scenario cuts gas demand by nearly twice as much (160 bcm in 2050), and the Accelerated Path to Net Zero scenario achieves slightly higher reduction (down to 141 bcm in 2050). Compared to the published 2024 energy transition scenarios by OIES (see Fulwood, 2024), the Baseline, High Carbon Price, and Low Carbon Price scenarios result in higher gas consumption by 2040 and 2050, suggesting a more substantial role of natural gas in decarbonisation than the 2024 OIES NZwthCCS and FRAG pathways for Europe. The NZwthCCS scenario (153 bcm in 2050) aligns more closely with the net-zero pathways modelled in this study, although the linear path to net zero results in higher demand in 2030 (+9%), 2040 (+19%) and marginally in 2050 (+4%), while the accelerated path to net zero reaches slightly lower level than NZwthCCS in 2050 (-8%). Meanwhile, the OIES 2024 DPS scenario (344 bcm in 2050) maintains gas consumption at levels comparable to the Low Carbon Price scenario, indicating a slower phase-out due to weaker decarbonising policy incentives. These comparisons highlight the impact of carbon pricing and climate policy stringency on gas demand and put the role of gas in the context of markets and policy development.

The elasticity of gas demand with respect to carbon prices provides insights into how a decarbonisation policy (via carbon price signals) influences fuel switching and consumption patterns. Table 4 presents the computed elasticities for the three carbon price scenarios in 2040 and 2050, showing two interesting insights from the modelling:

1. The impact of carbon pricing in 2040 on gas demand is more limited than in 2050 because of the relative availability of alternative energy supplies at competitive prices in 2050 compared to 2040;
2. The average elasticity of gas demand relative to changes in carbon prices across the scenarios is **low**, meaning that, for example, a 10% increase in carbon price results in only a potential reduction of gas demand in the range of 0.24% (high vs low scenarios in 2040) to 4.0% (High vs Baseline in 2050).

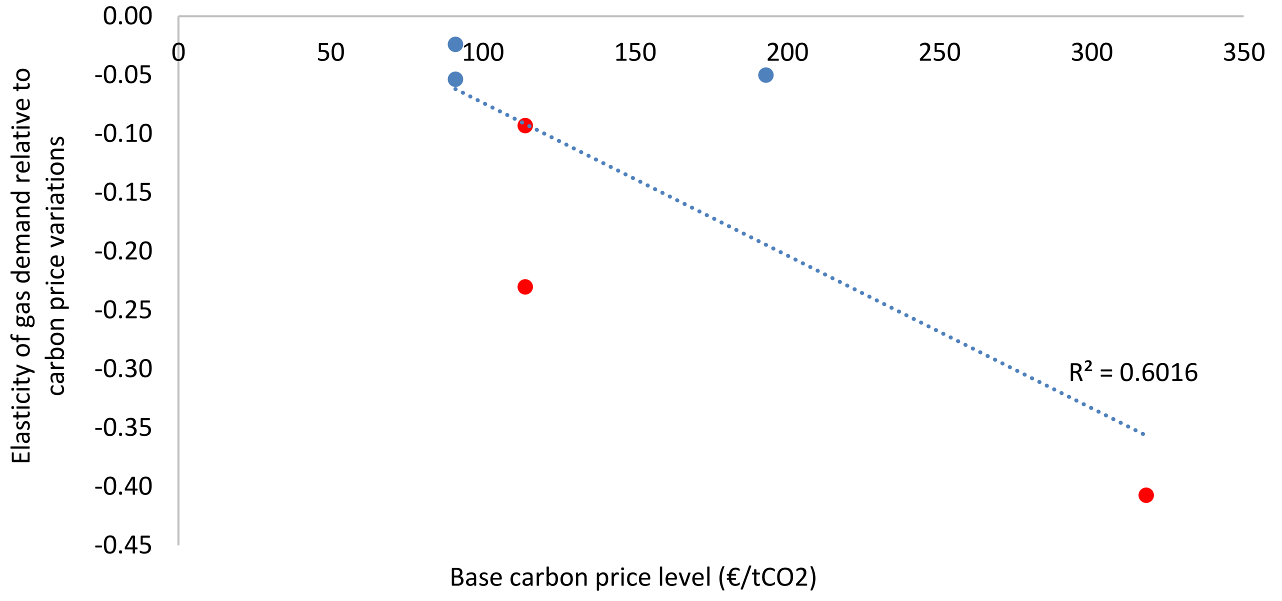
Table 4: Average gas demand elasticities relative to changes in carbon prices

Year	High vs Baseline	Low vs Baseline	High vs Low
2040	-0.051	-0.054	-0.024
2050	-0.404	-0.236	-0.094

Further, elasticity is not uniform across periods, as shown in the scatter plot (Figure 2: red dots represent 2050 elasticities, and blue dots represent 2040 elasticities). Considering only 2040 elasticities there is a much flatter trend between base carbon price levels and elasticities whereas in 2050 this relationship is steeper. Nevertheless, considering both 2040 and 2050, the fitted trend line suggests a relatively strong proportional relationship, where each increment in carbon price leads to progressively greater reductions in gas demand over time. The trend suggests that gas demand elasticity becomes more negative as carbon prices rise, indicating a stronger response to price increases at higher base carbon price levels, especially in 2050. As noted, in 2040, gas demand remains relatively inelastic, even under high carbon prices, due to infrastructure and technological constraints that slow the energy transition. By 2050, as decarbonisation technologies become relatively more mature and viable, the impact of higher carbon prices on gas demand becomes more pronounced relative to 2040.

The analysis suggests that while carbon pricing is a key driver of gas demand reduction, its impact is not linear across all price levels and years. The scatter plot confirms that gas demand remains relatively inelastic at lower carbon prices and in the near to medium term (by 2040), reflecting continued reliance on gas in final consumption sectors such as buildings, where switching costs remain high. However, the elasticity becomes more negative at higher carbon price levels in the longer term (2050). Despite this relative change between 2040 and 2050, these elasticities are considered **low** (less than 1 in absolute terms, i.e., **inelastic**), suggesting that decarbonisation will require much stronger policy support, sharper carbon price signals (see Table 2) and higher support costs. Importantly, low elasticities point to greater energy price volatilities as marginal energy costs depend on carbon and fuel costs.

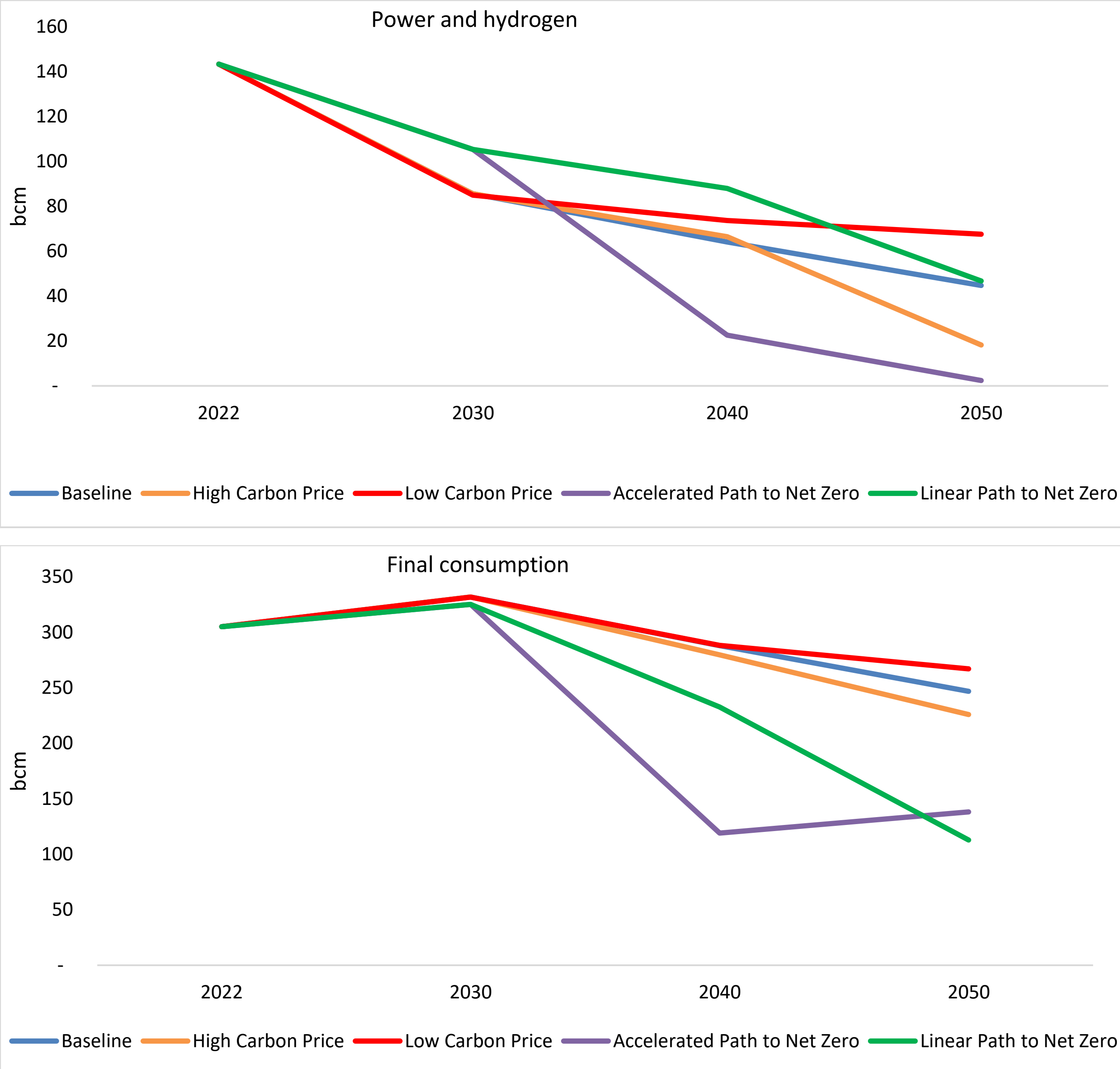
Figure 2: Base Carbon Price Level for ETS1 (€/tCO2) vs gas demand response



To further explore these dynamics, Figure 3 highlights sectoral gas demand evolution in power generation, hydrogen production, and final consumption. In particular, gas demand in power generation declines across all scenarios but at different rates, reflecting the stringency of decarbonisation policies modelled. In 2030, gas use in the transformation sector (power and hydrogen) is primarily for power generation, though hydrogen production begins to emerge, accounting for 3–9% of transformation gas demand. The Baseline and High Carbon Price scenarios show a 40% reduction in power sector gas use from 2022, while the Low Carbon Price scenario follows a similar trajectory with slightly higher gas use. In contrast, the Accelerated and Linear Path to Net Zero scenarios increase (relative to the other three carbon price scenarios) gas use to 105 bcm due to stronger policy signals favouring switching to gas in the power sector and kickstarting the production of blue hydrogen. By 2040, gas demand in power falls, with the Accelerated Path scenario dropping to 23 bcm. Meanwhile, the Baseline, High Carbon Price and Low Carbon Price scenarios show gas demand falling to only 64–74 bcm. Hydrogen's share of transformation gas grows significantly, reaching 50–51% in the Baseline and High Carbon Price cases.

By 2050, gas in power generation is nearly phased out in the most ambitious scenarios, while in others, hydrogen production absorbs a growing share of the remaining transformation gas. The Baseline scenario dedicates 60% of transformation gas to hydrogen, meaning only 18 bcm remains for power generation. The High Carbon Price scenario sharply reduces its reliance on gas for hydrogen (10%), indicating that a too-high carbon price in 2050 favours alternative hydrogen production methods. In contrast, the Low Carbon Price scenario continues to allocate 39% of transformation gas to hydrogen, keeping gas use in power generation higher than in other carbon price scenarios. The two paths to net zero see a limited role of blue hydrogen in 2050, potentially due to residual emissions.

Figure 3: Gas in power, hydrogen and final consumption sectors



Final gas consumption, including buildings, industry, transport, agriculture, and other uses, presents more complex dynamics. Similar to the trend observed in the net zero paths for power and hydrogen, gas demand initially rises in 2030 under all five scenarios modelled due to relatively strong policy signals (price-based and politically-driven targets) to switch to gas in end-use sectors. By 2040, reductions range from -5.6% (Baseline) to -24.0% (except for the aspirational policy scenario of a 90% reduction in 2040, which sees a reduction of 61%) relative to 2022, highlighting the higher cost of fuel-switching in final consumption than in power and hydrogen. The persistence of gas use in final consumption is particularly evident in the Low Carbon Price scenario, where demand remains high even in 2050 (267 bcm, -12.5% relative to 2022), reinforcing the findings on gas demand elasticity (Figure 2). The earlier analysis of price elasticities indicated that total gas demand remains relatively unresponsive to carbon prices in 2040, requiring more substantial policy intervention beyond carbon pricing alone.

The contrast between power/hydrogen and final consumption underscores the sectoral disparities in the impacts of policies on decarbonisation and transition speed. While power generation and hydrogen production respond more to policy interventions, final consumption gradually declines due to higher infrastructure, technological constraints and switching costs. This sectoral rigidity aligns with the observed pattern of gas demand elasticity, where gas use in buildings and industry is more resistant to price signals.

4. Conclusions

The analysis conducted using the Energy Futures Optimisation Model (EFOM) highlights the importance of understanding and quantifying costs and impacts of policy targets, measures and incentives. In particular, the research findings highlight the critical role of policy instruments – carbon pricing, renewable energy subsidies, and direct GHG caps – in shaping the decarbonisation trajectory, influencing fuel switching, investment in low-carbon technologies, and the cost implications of achieving climate targets.

A key finding from the modelling results is that achieving net zero emissions is exceptionally costly, requiring unprecedented financial and public support and breakthroughs in key energy technologies to reduce future abatement costs. Without such advancements, the marginal abatement cost (MAC) increases significantly as the system approaches full decarbonisation. The results suggest that alternative abatement strategies must be considered if Europe remains committed to keeping economic and industrial activity while meeting deep decarbonisation targets. These alternatives include increasing imports of low-carbon energy sources, such as hydrogen, bioenergy, and e-fuels, and integrating international carbon markets. Without these measures, the implicit MAC will remain prohibitively high as deep emissions cuts are reached.

Furthermore, two out of three carbon price scenarios demonstrate that achieving significant reductions in emissions at reasonable carbon price levels is possible with the modelled assumptions. Specifically, these scenarios achieve 84% and 86% reductions relative to 1990 levels, respectively, while keeping MAC within manageable ranges. However, this indicates that while strong carbon pricing can drive substantial emissions reductions, it alone is insufficient to achieve net zero emissions. Additional policy measures, financial support, technology breakthroughs, and alternative decarbonisation pathways are required to reach full decarbonisation targets. The role of gas in these scenarios, albeit smaller than now, remains not insignificant, as it serves as a transitional fuel, particularly in power generation and hydrogen production, supporting system flexibility while maintaining energy security.

Achieving net zero cost-effectively requires a balanced policy approach integrating carbon pricing, financial support for R&D and innovation, and alternative decarbonisation pathways, including low-carbon imports and linking with international carbon markets. Without these elements, the economic feasibility of deep decarbonisation is at risk due to high abatement costs.

The modelling work was primarily carried out to demonstrate the model's key capabilities. Limitations exist, and future work is needed to address them. In particular:

- The role of firm energy resources in providing energy security to hedge against low-probability high-impact events (such as renewable energy supply droughts combined with geopolitical risks of supply chain disruptions, etc.)
- Explicitly modelling the role of low-carbon energy imports and international carbon markets in achieving deep decarbonisation targets.
- Assessing the role of carbon capture, utilisation, and storage (CCUS) strategies, including the availability and scalability of direct air capture, CO2 storage, and CO2 utilisation, which could help reduce MAC under stringent climate targets.

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Appendix 1 – Model description

This section provides a high-level description of EFOM, while a detailed mathematical formulation of the model is outlined in Chyong et al. (2024). Figure A. 1 outlines the high-level structure of EFOM, with some further technical information. The energy system model is a partial equilibrium, linear programming optimisation model capable of a detailed representation of a present and future energy system. (For key supply and demand technologies modelled, see Figure A. 2). It is an economic optimisation model. Its objective is to minimise total energy system costs, comprising capital and operational costs, while meeting exogenously defined (projected) end-user demand for energy services, GHG emissions, and other constraints specified by the user (Figure A. 1).

Figure A. 1: High-level description of EFOM

Objective

- **Minimise total energy system costs:**
  - CAPEX of technologies
  - Variable opex (e.g., energy costs, variable O&M)
  - Infrastructure costs (e.g., gas, H_2, power transmission & distribution, storages)
  - Load shedding costs

Decision variables

- Capacity of conversion technologies, transport & storage infrastructure
- Hourly dispatch of energy, including inter-zonal flows

Constraints

- Hourly demand and supply balance constraint
- Techno-economic constraints, such as:
  - Power generation & P2X ramping limits,
  - Storage constraints,
  - EV charging constraints,
  - Inter-zonal transmission limits,
  - National transmission and distribution networks limits.

Scope

- Energy services demand in residential, commercial and agriculture
- Energy services demand in the road transport sector
- Energy demand in aviation, rail and inland navigation
- Energy demand in the industrial sector
- non-CO_2 emissions in aggriculture and other sectors

The model represents 27 regions, allowing endogenous trade in primary energy (Table A. 1). The model covers hourly dispatch and operations of energy conversion technologies and investment in power generation capacity, end-use low-temperature heat technologies, low-carbon hydrogen production, electricity- and hydrogen-based fuels production (e-fuels), end-use road transport technologies (e.g., internal combustion engine cars, battery-electric cars, etc.), storage and networks of methane (CH_4), hydrogen (H_2), electricity and CO_2. The model covers the final consumption sectors – residential, commercial, transport, and industry. The aggregated final consumption sectors in the model are as follows:

- The buildings sector represents the final energy services demand of residential, commercial and energy use in the agriculture sectors;
- Road transport represents the final energy services demand for road activities of passenger cars, public road transport (e.g., projected million passenger kilometres) and heavy goods vehicles (HGV) (e.g., projected million kilometre tonnes);
- Industry represents final energy consumption in the industrial sector;
- Other forms of transport represent final energy consumption through aviation, inland navigation, and rail transport activities.

In terms of supply and transformation technologies, the model takes into account:

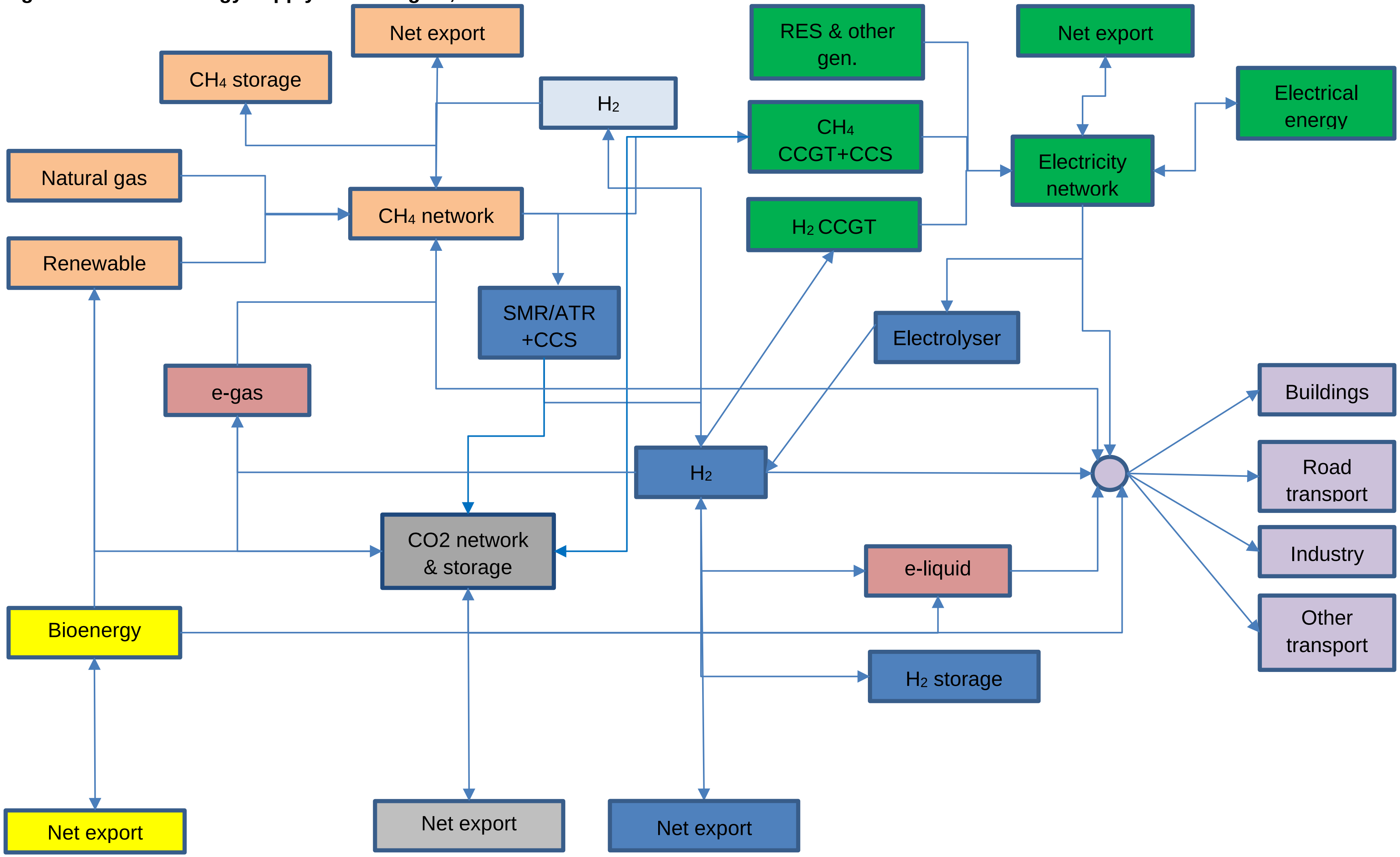
- Primary energy supply: crude oil, coal, uranium, biomass, natural gas (fossil origin), biomethane (renewable gas), renewable electricity from hydroelectric, wind, solar, geothermal, marine, etc.
- The natural gas sector includes pipeline transport for natural gas and biomethane and LNG via ships as well as underground storage;
- Secondary energy supply: diesel, gasoline, hydrogen from electrolysis and methane reformation, electricity from thermal generation, and synthetic electrofuels (e-gas and e-liquid).

- power generation (e.g., CCGT CH4/H2, nuclear, bioenergy, hydropower and other renewables, etc.) and storage technologies (e.g., electricity battery storage, hydro pumped storage, etc.) for the electricity sector;
- cross-border trade in energy (primary and derived), including via electricity transmission, CH4 and H2 pipelines;
- end-use technologies and energies in buildings and transport sectors.

Table A. 1: Spatial resolution and aggregation in EFOM

Regions in the model	Countries & Comments
UK	United Kingdom (UK)
Ireland	Republic of Ireland
Nordic	Norway (NO), Sweden (SE), Finland (FI), Denmark (DK), pipeline export & import
BE	Belgium (BE), Luxembourg (LU)
DE	Germany (DE)
NL	Netherlands (NL)
FR	France (FR)
IT	Italy (IT)
Baltics	Lithuania (LT), Latvia (LV), Estonia (EE)
PL	Poland (PL)
Eastern Europe	Czech Rep (CZ), Slovakia (SK), Hungary (HU)
Central Europe	Austria (AT), Switzerland (CH), Slovenia (SL)
South East Europe	Bulgaria (BG), Greece (GR), Croatia (HR), Romania (RO), Malta (MT), Cyprus (CY)
Iberia	Spain (ES), Portugal (PT)
North Africa	Utility-scale solar generation, hydrogen, e-gas, pipeline and LNG export
North Sea	Offshore wind generation
Sub-Saharan Africa	LNG export
Russia	Pipeline gas and LNG export
Caspian region	Pipeline gas export
North America	LNG export
Australia	LNG export
Middle East	LNG export
China	LNG import
Other Asia Pacific	LNG import
India	LNG import
Japan & Korea	LNG import
Rest of World	LNG import

Figure A. 2: Main energy supply technologies, networks and demand in the model



Appendix 2 – Net zero (linear) scenario: Data Inputs and Assumptions

This Appendix outlines technologies and energy vectors considered in this model version, the spatial resolution of the model, data calculations, processing and assumptions used in the modelling and model calibration to 2030 and 2050 policy targets. It covers the following:

1. Methods and assumptions used to derive input data from the EC LTS 1.5 TECH 2050 scenario and EC MIX2030;
2. Sources for techno-economic parameters for modelling (e.g., ramp rate, efficiency of power stations, etc.);
3. Other supply and demand projections.

This Appendix focuses on data inputs and assumptions for the net zero pathway (linear pathway). Assumptions and inputs for the other scenarios considered in this paper, which are different from the net zero pathway (linear pathway), are discussed in Appendix 3.

A.2.1 Model calibration to 2030

The modelling time horizon starts in 2030; therefore, the initial installed conversion, storage and network capacity are based on modelling results for the 2025 MIX scenario and NG FES for the GB market. Not all technologies are available and reported in the MIX scenario for the EU countries. For example, most power generation technologies at the EU country level were reported. Still, no information about gas infrastructure (network and storage capacity) or end-use space heating technologies is provided. The rest of this section outlines approaches and assumptions taken to compute the initial installed capacity of all infrastructure in the model. The section also outlines constraints and assumptions to calibrate the 2030 base year to the 2030 MIX results.

A.2.1.1 Power generation

The initial installed capacity for each technology and region in the model is taken from the MIX scenario 2025 for the EU countries and from the five-year forecast by NG FES (2023) for GB. To ensure consistency with the 2030 MIX scenario, an upper bound for gas CCGTs was implemented for 2030. The CCGT 2030 upper bound is based on MIX2030 country-level results, and for the GB market, it is based on the NG FES 2028 five-year forecast.

A.2.1.2 Buildings sector

MIX scenario results do not provide information about the end-use technology mix in the buildings sector, particularly regarding space heating technologies modelled here. Thus, the model was first run for 2030, assuming only gas boiler technology is available. This installed gas boiler capacity (1119 GW) was the initial value for optimising the 2030-50 pathways.

A.2.1.3 Road transport

The MIX scenario does not report vehicle stock by engine type. The EC LTS 2030 baseline (less ambitious than the MIX 2030 scenario) was used to compute the initial stock of road transport (Table A. 2). The initial values are given in the model as a starting point. The model optimises stock net addition according to our relative costs, techno-economic performance assumptions and emissions constraints.

Table A. 2: 2030 Initial road transport stock by engine type

	ICE gasoline	ICE diesel	ICE gaseous	Plug-in hybrid	Electric	Fuel cell
Passenger cars	40%	41%	5%	6%	8%	0%
Light-duty trucks	4%	84%	0%	6%	6%	0%
Heavy-duty trucks	0%	81%	5%	14%	0%	0%

Source: EC LTS 2030 Baseline modelling results

A.2.1.4 Networks

The installed capacity of energy networks was not reported in the MIX 2030 scenario. We, therefore, run the model to determine the optimal installed capacity for gas and electricity networks at the regional level and then use these values as shares and the 2019 peak hour demand to compute initial installed capacity. It is implicitly assumed that the initial installed capacity can at least deliver 2019 peak hour gas and electricity demand.

A.2.1.5 Storage

Existing (hydro-pumped storage capacity in 2020 and gas storage capacity in 2022) storage capacity was considered as an initial storage capacity for 2030 (Table A. 3)

Table A. 3: 2030 Initial storage capacity (GWh)

	Hydro Pumped storage	LNG storage	Underground gas storage
Baltics	7	3913	28987
BE	10	3834	7556
CentEurope	80		91933
DE	75	3333	269611
EastEurope	17		147367
FR	40	10281	132611
GB	32	14192	16000
Iberia	68	25108	37333
Ireland	2		
IT.	58	5525	196544
NL		5962	155189
Nordic		2209	11375
PL	14	4474	35689
SEE	15	4624	39667

Source: hydro pumped storage is taken from ENTSO-e; gas storage is taken from Chyong (2024[10])

A.2.2 Demand

Assumptions and data sources for the 2050 NZ are based on Chyong et al. (2024). This section, therefore, briefly summarises the main assumptions to compute values for 2030-50.

A.2.2.1 The buildings sector

The demand for thermal energy services in buildings is based on the EC JRC TIMES input database (Nijs, 2019[11]) for every country and region in our model. The following thermal energy services demand categories were considered for residential and commercial buildings:

1. Cooking thermal energy services demand.
2. Cooling thermal energy services demand.
3. Space heating thermal energy services demand.

[10] https://www.oxfordenergy.org/publications/burning-the-bridge-to-ostpolitik-stress-testing-europes-shift-from-russian-gas-to-renewables-using-a-global-energy-model/

[11] Wouter Nijs. (2019). JRC-EU-TIMES - JRC TIMES energy system model for the EU (1.1.1) [Data set]. Zenodo. https://doi.org/10.5281/zenodo.3544900

4. Water heating thermal energy services demand.
5. Lighting and appliances and specific electricity use demand.

Cooking, cooling, lighting and appliances are supplied by electricity. End-use technologies to meet these demand services were not in the scope of the current model. A choice of end-use technologies (e.g., gas boilers or heat pumps) to meet the demand for thermal energy services for space and water heating in the buildings sector (residential and commercial buildings) is optimised in the model. Thermal energy services demand is assumed to remain unchanged throughout the modelling horizon. Therefore, the effects of investments in energy efficiency (e.g., buildings retrofits) are left for future model improvements.

Further, agriculture energy demand has been included in the 'buildings' sector's node, following the EC PRIMES modelling convention. In the modelling, end-use technology options for agriculture's final energy demand (e.g., different farming machine drives) are not explicitly considered, so a historical fuel mix from the EC JRC TIMES model (2010) with the following assumptions was used:

1. Diesel will be replaced by carbon-neutral e-liquid gradually from 2030 to 2050;
2. Natural gas will be replaced by carbon-neutral biomethane gradually from 2030 to 2050
3. Coal will be replaced by biomass gradually from 2030 to 2050

Note that agricultural energy demand is treated separately from the buildings sector's demand for space and water heating services.

A.2.2.2 Transport

The demand projection for road transport activities for every country in the model for the 2050 Net Zero (NZ linear) scenario is based on Chyong et al. (2024). The demand projection for 2030-2045 is based on interpolation using 2015's actual transport activities (taken from the EU Reference scenario 2016) and 2050's NZ (1.5 TECH scenario) projection. Following Chyong et al. (2024), other transport modes, including aviation, rail and inland navigation, were not modelled endogenously in this model version. Hence, the final energy consumption from these other transport modes is calibrated to EC 2050 NZ (1.5 TECH scenario) (taken from Chyong et al., 2024), and 2015 data is used to interpolate demand for 2030-45. Table A. 4 outlines all other transport modes' projected final energy consumption.

Table A. 4: Fuel mix in the non-road transport sector (mtoe) in Europe

	2030	2035	2040	2045	2050
Diesel	57.30	49.22	41.14	33.06	24.99
E-liquids	9.59	12.78	15.98	19.18	22.37
Biomass	6.96	9.29	11.62	13.94	16.27
Electricity	2.60	4.73	6.87	9.01	11.14
Hydrogen	0.06	0.07	0.09	0.11	0.13
E-gas	0.14	0.18	0.23	0.28	0.32
Biomethane	0.03	0.04	0.05	0.06	0.07

A.2.2.3 Industry

Final energy demand in the industrial sector was calibrated to the results of EC 2030MIX and 2050 NZ (the 2050 NZ was based on the results of Chyong et al., 2024) (see Table A. 5). Note that "other" fuels are not modelled and is reported for completeness.

Table A. 5: Fuel mix in the industrial sector (mtoe) in Europe

	Electricity	Natural gas	Biomethane	Hydrogen	E-gas	Biomass	Other	Total
2030	89	83	6	0	0	21	108	307
2035	96	50	7	7	3	22	89	274
2040	104	35	8	15	5	23	70	259
2045	111	19	9	22	8	24	51	244
2050	119	4	10	29	11	25	32	230

Sources: own calculations based on EC MIX2030 (for 2030) and Chyong et al. (2024) for NZ. 2050; other years were interpolated using 2030 and 2050

A.2.3 Supply, conversion and power generation

All costs and techno-economic parameters related to the supply of primary energy, conversion to secondary energy and power generation technologies are based on the dataset reported in detail in Chyong et al. (2024) for both 2030 and 2050. Values for the intermediate years were interpolated using 2030 and 2050 parameters. All costs are in 2023 Euros.

A.2.4 Networks

All costs and techno-economic parameters related to the operation and expansion of electricity, natural gas, and hydrogen networks (cross-border and national networks) are based on the dataset Chyong et al. (2024) reported for 2030 and 2050. Values for the intermediate years were interpolated using 2030 and 2050 parameters. All costs are in 2023 Euros.

A.2.5 Storage

All costs and techno-economic parameters related to the operation and expansion of electricity, natural gas, hydrogen, and CO2 storage are based on the dataset Chyong et al. (2024) reported for 2030 and 2050. Values for the intermediate years were interpolated using 2030 and 2050 parameters. CO2 storage variable and capital costs were updated using data from Holz et al. (2021[12]). Natural gas variable and capital costs used in this modelling study were updated using Chyong (2024[13]) and Ramboll (2008[14]) data. All costs are in 2023 Euros.

A.2.6. Buildings end-use heat technologies

The costs of heat pumps and hybrid heat pumps (Table A. 6) are updated based on data from Chyong et al. (2024) and Element Energy (2017[15]). All costs are in 2023 Euros.

Table A. 6: Capital cost of heat technologies (Eur/KW.)

	2030	2050
Boilers condensing gas	195.0	210.0
Heat pump air source	1114.7	1200.5
Hybrid heat pump	716.6	771.7
Boilers condensing H_2	195.0	210.0

Source: own calculations based on data from Chyong et al. (2024) and from Element Energy (2017)

[12] https://www.sciencedirect.com/science/article/pii/S0140988321004941
[13] https://www.oxfordenergy.org/publications/burning-the-bridge-to-ostpolitik-stress-testing-europes-shift-from-russian-gas-to-renewables-using-a-global-energy-model/
[14] Ramboll (2008). “Study on Natural Gas storage in the EU” Ramboll Oil & Gas, Virum, Denmark, October 2008
[15] https://assets.publishing.service.gov.uk/media/5ad6142040f0b617dca714c5/Hybrid_heat_pumps_Final_report-.pdf

A.2.7 Road transport technologies

Costs and techno-economic data for road transport technologies were taken from Chyong et al. (2024).

A.2.8. GHG emissions

Three types of CO_2 emissions constraints are implemented in the model: (1) economy-wide net CO_2 emissions (column 3, Table A. 7), (2) buildings-sector emissions (column 1, Table A. 7), and (3) road transport-sector emissions (column 2, Table A. 7). These constraints guide emissions reductions towards achieving Net Zero in 2050.

Table A. 7: GHG emissions trajectories (mtCO2e) for Europe

	Buildings sector	Road transport	Total emissions
	[1]	[2]	[3]
2030	335	669	2,622
2035	259	505	2,014
2040	183	340	1,407
2045	107	176	799
2050	31	11	0

Sources: own calculations based on EC LTS (NZ 2050)

A.2.9 European-level constraints

This section details the upper bounds for key electricity generation technologies to reflect resource endowments (e.g., availability of space for wind and solar) and calibration to the EC's 2030 MIX and 2050 NZ scenario results. Therefore, at the European level, the EC's 2030 MIX[16] and 2050 NZ are used to interpolate for the years between 2030 and 2050 (Table A. 8).

Table A. 8: Upper bound for key electricity generation technologies for Europe (GW)

	2030	2035	2040	2045	2050
Biomass CCS	1	13	25	37	49
Nuclear	108	112	115	118	121
Wind Onshore	382	476	570	665	759
Wind Offshore	94	184	273	362	451
Solar PV*	426	604	782	960	1138
Tidal and Wave	1	3	6	9	12
Geothermal	2	3	3	4	5
Hydro	150	169	189	208	228
Fossil fuel (CCS)	1	5	9	13	17

Source: 2050 is based on the EC LTS NZ 2050 scenario, while 2030 is based on the EC 2030 MIX scenario and a five-year forecast for the GB based on NG ESO 2023; the years between 2030 and 2050 are linear interpolations
Notes: * includes EU27, UK, NO, CH, and North Africa's solar potential export capacity to Europe

[16] And for GB the five-year forecast (2028) by the NG ESO (2023)

Permanent underground CO_2 storage capacity is based on the recent EC (2024[17]) industrial carbon management strategy. In particular, storage capacity will increase from 50 mtCO2 in 2030 to 250 mtCO2 by 2050. In addition, the UK's upper bound on CO2 storage capacity will likely be 25 mtCO2 in 2030, increasing to 75 and 180 mtCO2 by 2050. An average from the 2050 UK CO2 storage capacity range is assumed, giving 75 mtCO2 in 2030 to 377.5 mtCO2 in 2050 for the EU plus the UK.

A.2.10 Regional and country-specific constraints

National-level upper bounds are developed and applied for most conversion technologies. This section outlines the methodology used to compute upper bounds for onshore wind technology, followed by offshore wind and other technologies subject to capacity expansion constraints.

Table A. 9: Upper bound for onshore wind capacity expansion (GW)

Country/region	2030	2035	2040	2045	2050
BE	8.6	15.2	19.5	23.6	26.5
DE	91.7	127.6	206.9	284.6	462.5
FR	40.4	66.6	95.7	116.8	170.4
IT	15.9	19.1	34.3	62.2	89.6
GB	24.4	40.0	70.2	101.3	165.9
Ireland	8.0	12.6	15.7	14.9	20.7
Nordic	32. 4	51.1	62.3	55.5	76.7
NL	18.5	36.8	44.7	51.7	48.9
Baltics	3.0	4.4	6.5	8.3	12.2
PL	17.7	29.1	39.6	43.3	63.0
EastEurope	8.3	12.3	18.8	25.3	38.2
CentEurope	8.3	12.4	27.5	42.7	81.3
SEE	23.8	40.5	49.1	40.3	57.0
Iberia	43.9	49.7	91.4	172.9	247.3

The methodology for determining upper-bound capacity expansion for onshore wind relies on multiple data sources to establish historical growth trends and projections. Three key datasets inform this approach:

1. The Energy Institute's (EI) Statistical Review of World Energy (2024) provides historical trends in wind capacity deployment across European countries and regions (Energy Institute, 2024[18]). Historical deployment data from this source is used to compute each region's 10-year historical average compound annual growth rate (CAGR). This CAGR is applied to project upper bounds for 2025, 2030, and 2035, ensuring that near-term expansion projections remain aligned with observed historical trends.

2. European Commission's JRC ENSPRESO dataset on renewable energy potential provides technical estimates for onshore wind across Europe (Ruiz Castello et al., 2019[19]). This dataset provides insights into the 2050 technical potential for each region and country within the model.

[17] https://energy.ec.europa.eu/document/download/6b89e732-fea4-480b-9d2e-cf64de90247e_en?filename=Communication_-_Industrial_Carbon_Management.pdf
[18] Energy Institute (2024). *Statistical Review of World Energy 2024*. Energy Institute. Available at: https://www.energyinst.org/statistical-review
[19] Ruiz Castello, P., et al. (2019). *ENSPRESO – an open, EU-28 wide, transparent and coherent database of wind, solar, and biomass energy potentials*. JRC Technical Reports. European Commission. Available at: https://publications.jrc.ec.europa.eu/repository/handle/JRC116900

3. The European Commission's 2030 MIX Scenario (European Commission, 2023) is used to compute an expansion slope, which is then modified and incorporated into the analysis.

Two distinct methodologies are used to compute the 2050 upper bound, ensuring that near-term growth remains consistent with historical trends while the long-term trajectory reflects a "realistically ambitious" capacity expansion.

The "Enspresso" Approach

For certain countries and regions (Germany, Belgium, the Netherlands, the UK, and Central Europe), the 2050 upper bound was directly sourced from the Enspresso dataset, reflecting full utilisation of technical potential and policy-driven expansion pathways. In these cases:

- CAGR-derived projections were used for 2025, 2030, and 2035, ensuring near-term capacity expansion aligns with historical trends.
- Interpolation was applied for 2040 and 2045, ensuring a gradual capacity expansion trajectory between the CAGR-based projections for 2025 and the fixed 2050 Enspresso upper bound.

The "MIX/Hybrid" Approach

For several countries and regions, using Enspresso's 2050 technical potential would lead to unrealistic deployment trajectories, as their installed base in 2025 is too low to support such high expansion rates.

In these cases, a MIX/Hybrid approach was applied, where:

- The 2030-2050 trajectory was computed using a modified slope derived from the EC MIX scenario (2020-2030 trends).
- The modified slope was adjusted only for 2045 and 2050 to ensure a more ambitious potential than the slope derived solely from the 2020–2030 MIX scenario, reflecting a higher rate of capacity expansion post-2030.

This approach ensures that capacity expansion remains consistent with policy-driven trajectories while allowing for moderate acceleration beyond 2030 and prevents overly steep deployment rates that would be unrealistic given the low starting capacity in 2025.

Computing the upper bound for offshore wind capacity expansion (Table A. 10) follows the same methodology applied to the onshore wind. With a few differences:

- Unlike onshore wind, several European countries had no offshore wind capacity as of 2023. This means that historic CAGR could not be applied to project their expansion. For these countries (Italy, the Baltics, and Southeast Europe (SEE)), the MIX capacity projection for 2030 was used as the starting point, and the historical CAGR (for onshore wind) was then applied to project the 2035 upper bound. Then, the MIX/Hybrid method was applied to derive the entire trajectory for 2050, following the same modified slope approach used in onshore wind expansion.
- For Belgium, the Enspresso value for 2050 (1.85 GW) appeared too low compared to the MIX2030 projection for 2030 (5.11 GW). Since it would not be realistic for Belgium's offshore wind capacity to decrease post-2030, the MIX2030 projection 2030 (5.11 GW) was used as Belgium's 2050 upper bound instead of the Enspresso value. Linear interpolation was then applied to compute values for 2030-2045 using the 2025 MIX scenario and the new 2050 upper bound.

For onshore and offshore wind, the sum of all regional upper bounds in 2050 was allowed to exceed the EC 1.5 TECH 2050 (see Table A. 8). This ensures that the overall capacity expansion aligns with the EC 1.5 TECH trajectory, but the model retains the flexibility to optimise regional and country-level deployment. This additional headroom allows the optimisation framework to adjust national-level capacity allocation based on resource availability, economic and policy feasibility, and various system integration constraints.

Table A. 10: Upper bound for offshore wind capacity expansion (GW)

Country/region	2030	2035	2040	2045	2050
BE	3.1	3.6	4.1	4.6	5.1
DE	20.1	28.0	46.2	64.1	105.6
FR	4.4	7.2	22.8	60.7	80.5
IT	1.9	2.2	13.5	42.3	56.3
GB	35.8	58.6	73.5	89.8	96.5
Ireland	1.2	1.8	7.0	12.2	26.7
Nordic	3.5	5.6	11.4	24.3	31.7
NL	9.7	19.3	37.7	55.6	96.8
Baltics	0.5	0.8	3.9	11.6	15.5
PL	1.0	1.7	6.7	11.7	25.7
SEE	0.7	1.2	5.5	16.0	21.3
Iberia	0.1	0.1	7.1	14.1	35.2

The methodology for defining upper bounds on solar photovoltaic (PV) capacity expansion is structured similarly to wind technologies but incorporates solar-specific deployment patterns, including historical installation rates, rooftop and utility-scale segmentation, and national-level variations (see Table A. 11). The following approaches were used to determine the 2050 upper bound for solar PV, depending on the availability of historical deployment data.

1. "Highest Built Rate Historic" Approach. For most countries, historical annual absolute growth rates in solar PV installations over the past 10 years were obtained from the Energy Institute's Statistical Review of World Energy (2024) (Energy Institute, 2024). The highest year-on-year absolute capacity growth observed in the past decade was assumed to persist until 2050 to set an ambitious but historically plausible build rate. The starting point for projections was the 2025 solar PV capacity from the MIX scenario. 2030, 2035, 2040, 2045, and 2050 projections were computed by applying the highest historic absolute annual increase to each time step.

2. "Enspresso-Based" Approach. For countries with negligible solar PV installations in 2023 or unavailable historical growth rates, the Enspresso technical potential was used as the 2050 upper bound. The 2025 value from the MIX scenario served as the starting point. A linear interpolation was applied to compute intermediate values for 2030, 2035, 2040, and 2045, ensuring a smooth trajectory toward the 2050 upper bound. This approach was applied to Ireland, the Baltic region, and Poland, where historical installation data was insufficient for calculating absolute annual growth rates.

3. For North Africa's utility-scale solar PV upper-bound export-oriented capacity to Europe, twice the highest historic annual solar PV growth rate is applied to account for the region's high solar potential.

Further, solar PV is deployed in two distinct segments: residential rooftop solar PV – typically installed at smaller scales, driven by consumer incentives and self-consumption economics, and utility-scale solar PV – large-scale installations feeding directly into the grid, requiring dedicated land and infrastructure. Historical national data on the split between rooftop and utility-scale solar PV was used to proportionally distribute the total upper bound for each country/region into these two categories.

As with wind capacity expansion, the sum of country and regional solar PV upper bounds in 2050 was allowed to exceed the EC 1.5 TECH total for 2050 (see Table A. 8). This ensures that the model retains the flexibility to optimise regional capacity deployment while meeting the overall 2050 target for solar PV expansion in the European energy transition pathway.

Table A. 11: Upper bound for solar PV capacity expansion (GW)

Country/region	2030	2035	2040	2045	2050
GB	34.6	56.5	78.4	100.3	122.2
Ireland	1.2	2.3	3.4	4.5	5.6
Nordic	6.9	8.1	9.4	10.7	11.9
BE	13.6	18.7	23.8	28.9	34.0
DE	98.6	135.1	171.7	208.3	244.9
NL	29.8	38.9	48.0	57.0	66.1
FR	34.7	40.7	46.7	52.8	58.8
IT	85.1	135.4	185.6	235.9	286.2
Baltics	4.8	8.8	12.8	16.7	20.7
PL	20.3	28.4	36.5	44.6	52.7
EastEurope	14.8	20.5	26.3	32.1	37.9
CentEurope	4.5	5.5	6.6	7.6	8.7
SEE	24.7	32.4	40.1	47.8	55.5
Iberia	55.3	77.0	98.7	120.4	142.0
North Africa*	21.6	43.2	64.8	86.4	108.0

Notes: *North Africa's solar potential export capacity to Europe

Table A. 12 outlines the upper bounds for biomass-based power generation capacity potentials. These upper bounds were determined using a combination of projected growth rates from policy scenarios (MIX2030), historical coal-fired generation capacity, and considerations for retrofitting coal plants to operate on bioenergy.

First, the biomass capacity projections for 2025 and 2030 were taken from the EC MIX scenario, and the growth rate between these two years was computed. This growth rate was then applied as a constant factor to extrapolate the upper bounds for 2035, 2040, 2045, and 2050. This method was used as the default approach, ensuring that biomass capacity expansion follows a trajectory consistent with policy-driven expectations in the near term while allowing for continued expansion beyond 2030 as deep decarbonisation pathways will likely require bioenergy as a key resource.

In cases where the computed growth rate appeared too high (only Ireland has a very high growth rate from MIX2030) – significantly deviating from the general trend observed across all regions – it was considered an outlier. Instead of using this high growth rate to project capacity expansion to 2050, an alternative approach was applied. Specifically, the total coal-fired power generation capacity in 2020 for that country or region was identified, and this capacity was doubled to define the 2050 biomass capacity upper bound. This adjustment reflects the possibility of converting or repurposing coal-fired power plants for biomass generation.

A similar adjustment was made for regions where the computed growth rate between 2025 and 2030 was negative (Germany and Iberia region), indicating a decline in projected biomass capacity under the MIX2030 scenario. The same alternative method was applied since a negative growth trajectory does not align with the broader long-term trend of maintaining biomass generation as part of the transition to deep reductions in emissions. The 2020 coal-fired generation capacity was identified to establish the 2050 biomass capacity upper bound, again reflecting the assumption that a share of existing coal infrastructure could be converted to biomass-fired generation.

Table A. 12: Upper bound for biomass-based power generation capacity expansion (GW)

Country/region	2030	2035	2040	2045	2050
GB	3.5	3.5	3.5	3.5	3.5
Baltics	1.0	1.2	1.4	1.6	1.8
BE	1.5	2.0	2.7	3.5	4.6
CentEurope	2.9	2.9	3.0	3.0	3.9
DE	14	21.2	28.2	35.2	42.2
EastEurope	2.3	3.1	4.0	5.3	7.0
FR	6.4	6.9	7.6	8.3	9.1
Iberia	5.0	6.3	7.7	9.0	10.4
Ireland	0.5	0.9	1.4	1.8	2.2
IT	11.7	13.5	15.7	18.2	21.0
NL	1.9	2.4	2.9	3.5	5.1
Nordic	12.3	13.5	14.9	16.4	18.0
PL	2.1	2.1	2.2	2.2	3.3
SEE	2.0	2.3	2.7	3.2	3.7

Table A. 13 outlines the upper bound for hydroelectric generation capacity expansion implemented in the model. It is based on the historical distribution of generation capacity by countries and regions (taken from ENTSO-e), adjusting for the growth rate of the EC's 1.5 TECH scenario (see Table A. 8).

Table A. 13: Upper bound for hydroelectric power generation capacity expansion (GW)

Country/region	2030	2035	2040	2045	2050
Baltics	1.8	1.9	2.9	3.1	5.0
BE	0.2	0.2	0.3	0.3	0.3
CentEurope	16.1	17.6	26.4	28.5	30.6
DE	6.2	6.8	10.6	11.5	13.4
EastEurope	3.0	3.3	4.8	5.2	8.3
FR	24.4	26.7	29.1	31.4	33.8
Iberia	25.0	27.4	29.8	32.3	34.7
Ireland	0.3	0.3	0.4	0.5	0.5
IT	19.6	21.5	23.4	25.3	27.2
NL	0.0	0.0	0.1	0.1	0.1
Nordic	50.4	55.2	60.1	65.0	69.8
PL	1.1	1.2	1.8	1.9	3.0
SEE	16.5	18.0	19.6	21.2	22.8

The upper bounds for nuclear power capacity expansion (Table A. 14) were determined based on a review of national energy policies[20], planned capacity additions, phase-out decisions, and the EU-wide 1.5 TECH scenario. In particular, for Belgium and Iberia, the upper bounds reflect national policy decisions to extend existing nuclear capacity only until 2035, after which nuclear power is phased out. As a result, nuclear capacity remains constant between 2030 and 2035 but drops to zero by 2040.

[20] Mostly based on information published by the World Nuclear Association (https://world-nuclear.org/)

Table A. 14: Upper bound for nuclear power generation capacity expansion (GW)

Country/region	2030	2035	2040	2045	2050
BE	3.9	3.9	0.0	0.0	0.0
CentEurope	3.7	3.7	1.9	1.9	1.9
EastEurope	11.1	12.3	13.5	14.7	15.8
FR	63.0	64.2	65.4	66.5	68.7
Iberia	7.1	7.1	0.0	0.0	0.0
NL	0.5	0.5	2.0	3.5	3.5
Nordic	12.3	12.3	14.8	14.8	14.8
SEE	4.7	4.7	8.7	8.7	8.7
GB	5.6	5.6	7.9	7.9	7.9

For Central Europe, the upper bound accounts for Switzerland's planned nuclear phase-out by 2035, leaving only Slovenia with nuclear capacity beyond that point. The projection assumes that 0.7 GW of remaining capacity in Slovenia post-2035 will be supplemented by 1.2 GW of planned nuclear capacity in Slovenia and Croatia.

Eastern Europe's nuclear capacity upper bound is derived by taking 50% of the year-on-year increment from the EU-wide nuclear capacity bound under the 1.5 TECH scenario (see Table A. 8). This assumption reflects expected expansions in nuclear capacity across Eastern Europe. A similar approach was applied to France, where its incremental upper bound is derived by taking the year-on-year increment from the 1.5 TECH scenario and an additional 1 GW by 2050 so that the total for all regions aligns with the EU-wide bound in 2050.

For the Netherlands, the upper bound reflects planned nuclear expansion, with 1.5 GW of new capacity added by 2035 and another 1.5 GW by 2040. The Nordic region's nuclear upper bound incorporates Sweden's planned expansion, with an additional 2.5 GW expected by 2035. For Southeast Europe, the upper bound reflects planned new nuclear capacity additions in Bulgaria and Romania. Finally, the upper bound for Great Britain follows the projections outlined in the UK's Future Energy Scenarios (FES).

Table A. 15 outlines our assumptions regarding the upper bound for capacity expansion of oil-, gas-, hydrogen- and coal-based power generation technologies in 2030. The reason for imposing the 2030 limits is to facilitate model calibration to 2030 MIX scenario results and to limit the infeasible expansion of immature technologies (e.g., hydrogen-based power generation). The 2025 MIX scenario results for oil and gas capacity are used as a baseline (for GB, the 5-year forecast in the FES scenario is applied), with a 50% uplift to provide flexibility for the model to deviate from the MIX scenario. The resulting values serve as upper bounds for 2030. Coal upper-bound capacity is capped at the 2025 level (from the MIX scenario) in 2030. Beyond 2030, no upper bound limits are imposed on these technologies, allowing the model to optimise capacity expansion without constraints.

Table A. 15: Upper bound for other generation capacity expansion (GW) in 2030

Country/region	Oil	Gas	H2	Coal
GB	0.2	50.3	0.0	0.0
Ireland	0.5	6.7	0.0	0.8
Nordic	2.8	8.6	0.0	3.4
BE	0.3	15.3	0.0	0.0
DE	2.2	30.5	0.0	38.0
NL	0.1	19.0	0.0	3.4
FR	2.7	9.3	0.0	0.1
IT	4.8	52.7	0.0	0.0
Baltics	0.1	3.3	0.0	1.6
PL	0.2	16.3	0.0	26.3
EastEurope	0.2	8.3	0.0	10.6
CentEurope	0.8	5.8	0.0	0.6
SEE	5.0	19.5	0.0	8.0
Iberia	5.9	47.5	0.0	3.9

Appendix 3 – Alternative Scenarios: Data Inputs and Assumptions

Five distinct scenarios are developed within the Energy Futures Optimisation Model (EFOM) to explore the role of policy instruments in decarbonising the European energy system. The objective of these scenarios is to evaluate their impact on the energy system, particularly on natural gas, low-carbon, and renewable fuels. While Appendix 2 gives a detailed account of the data and assumptions used to create **the net zero (linear pathway)** scenario, this Appendix summarises key assumptions behind the definitions of alternative scenarios: the Baseline, High and Low Carbon Price alternatives, and Accelerated Path to Net Zero. The Accelerated Path to Net Zero scenario follows the same structure as the Linear Path to Net Zero (see Appendix 2), with the exception that the model is constrained to achieve a 90% reduction in GHG emissions by 2040. The remainder of this Appendix focuses on the three carbon price scenarios.

To ensure consistency with established decarbonisation scenarios and account for the complexities of sector-specific interactions, the calibration of final energy demand and GHG emissions in the industrial, agriculture, and other (non-road) transport sectors are based on projections from the European Commission's (EC) long-term scenarios. This calibration ensures alignment between the three carbon price scenarios modelled in this paper with the EC's detailed modelling analysis while limiting the scope of this model and research (Table A. 16). Explicitly modelling the interactions within these sectors and emissions categories is highly complex. Industrial processes, agricultural non-CO_2 emissions, and land-use emissions involve detailed and complex interdependencies beyond this model's scope. Thus, calibrating these sectors to the EC's scenarios ensures that results remain robust without requiring detailed sectoral modelling.

Table A. 16: Mapping of carbon price scenarios to EC's long-term scenarios

Sectors	Baseline	High Carbon Price	Low Carbon Price
Industry (final demand & GHG emissions)	Average of EC's 80% GHG emissions reduction scenarios*	EC's 90% GHG reduction scenario ("Combo" scenario**)	EC's Baseline***
Non-road transport (final demand & GHG emissions)	Average of EC's 80% GHG emissions reduction scenarios	EC's 90% GHG reduction scenario ("Combo" scenario)	EC's Baseline
Non-CO2 Emissions in Agriculture	Average of EC's 80% GHG emissions reduction scenarios	EC's 90% GHG reduction scenario ("Combo" scenario)	EC's Baseline
Non-CO2 Emissions in other sectors	Average of EC's 80% GHG emissions reduction scenarios	EC's 90% GHG reduction scenario ("Combo" scenario)	EC's Baseline
LULUCF GHG Emissions	Average of EC's 80% GHG emissions reduction scenarios	EC's 90% GHG reduction scenario ("Combo" scenario)	EC's Baseline

Source: EC (2018[21])

Notes: * 80% GHG reductions excluding LULUCF and 85–88% reductions including LULUCF by 2050 compared to 1990 levels; ** 90% GHG reduction including LULUCF by 2050 compared to 1990 levels; *** 62% GHG reduction excluding LULUCF and 66% including LULUCF by 2050 compared to 1990 levels

The carbon price values in this study are calibrated to established projections from the European Commission's long-term scenarios and the UK Department for Energy Security and Net Zero (DESNZ). The assumed carbon price levels in the three scenarios – Baseline, High and Low Carbon Price – influence emissions reduction efforts across key sectors, particularly those covered under the EU Emissions Trading System (ETS1) and the expanded ETS2 system for buildings and transport (Table A. 17).

[21] https://climate.ec.europa.eu/document/download/dc751b7f-6bff-47eb-9535-32181f35607a_en?filename=com_2018_733_analysis_in_support_en.pdf

Carbon price level applied to the ETS1-covered sectors:

- **2030 Carbon price for all scenarios:** The 2030 carbon price in all three scenarios is taken from the 2030 MIX scenario in the EC's analysis. The EC's original assumption for 2030 was €44/tCO_2 for ETS sectors and €10/tCO_2 for non-CO_2 emissions, summing to €54/tCO_2 at 2015 prices. Adjusted to 2023 price levels, this results in a 2030 carbon price of €68/tCO_2, applied consistently across all scenarios modelled here.
- **Baseline Scenario** (Scenario 1): The 2050 carbon price is set at €318/tCO_2 (2023 prices), derived from the EC's long-term strategy scenarios targeting an 80% GHG reduction, where the carbon price was initially set at €250/tCO_2 at 2013 price levels.
- **High Carbon Price** Scenario (Scenario 2): The 2050 carbon price is €445/tCO_2 (2023 price level), aligning with the EC's net zero scenario (1.5 TECH), which assumes a carbon price of €350/tCO_2 at 2013 price levels.
- **Low Carbon Price** Scenario (Scenario 3): The 2050 carbon price for ETS1 is set at €114/tCO_2 (2023 price level), based on the DESNZ Carbon Price Forecast, specifically the "Low Sensitivity – High Fossil Fuel Prices and Low Economic Growth" scenario from the UK government's traded carbon values report (2023[22]).

Carbon price applied to the ETS2-covered sectors (buildings and road transport):

- The ETS2 carbon price remains fixed for all scenarios at €54/tCO_2 (2023 price level[23]) from 2030 to 2050. While the EC aims to fix the ETS2 price at this level for the first three years (2027-30), this price level is applied for the whole modelling period (to 2050), considering that existing national road tax charges (fuel excise duty rates, see Table A. 18) are also applied to petrol and diesel as environmental and climate policy in the road transport sector.

Table A. 17: Carbon prices (2023 prices) applied in the alternative scenarios.

		2030	2040	2050
ETS1	Baseline	68	193	318
	High carbon price	68	257	445
	Low carbon price	68	91	114
ETS2	All carbon price scenarios	54	54	54

Notes: these carbon price pathways are applied to ETS1 sectors, while for ETS2 sectors, a carbon price of €45/tCO2 (at 2020 prices) is assumed, consistent with the assumption by the EC[24]; the values in 2040 are linear interpolation using values for 2030 and 2050

Fuel excise duty rates for petrol and diesel were compiled from the Tax Foundation's 2023 report on fuel taxation in Europe, which provides comprehensive data on excise duties for gasoline and diesel across EU Member States (Hoffer, 2023[25]) as well as from the Norwegian Tax Administration (Skatteetaten, 2024[26]), and the Swiss Federal Customs Administration (Swiss Customs, 2024[27]).

[22] https://www.gov.uk/government/publications/traded-carbon-values-used-for-modelling-purposes-2023/traded-carbon-values-used-for-modelling-purposes-2023
[23] According to the EC if, during the first three years the ETS2 is operational, the price of allowances exceeds €45 (at 2020 prices, i.e. adjusted for inflation), additional allowances may be released from the ETS2 market stability reserve to address excessive price increases. Allowances may also be released from this reserve if the price of allowances increases too rapidly (https://climate.ec.europa.eu/eu-action/eu-emissions-trading-system-eu-ets/ets2-buildings-road-transport-and-additional-sectors_en)
[24] https://climate.ec.europa.eu/eu-action/eu-emissions-trading-system-eu-ets/ets2-buildings-road-transport-and-additional-sectors_en
[25] Hoffer, A. (2023, August 15). Diesel and Gas Taxes in Europe, 2023. Tax Foundation. Retrieved from https://taxfoundation.org/data/all/eu/gas-taxes-in-europe-2023/
[26] Norwegian Tax Administration (Skatteetaten). (2024). Road Usage Tax on Fuel. Retrieved from https://www.skatteetaten.no/en/rates/road-usage-tax-on-fuel/
[27] Swiss Federal Customs Administration. (2024). Petroleum and Fuel Excise Duties. Retrieved from https://www.bazg.admin.ch/bazg/en/home/informationen-firmen/inland-abgaben/mineraloelsteuer.html

Table A. 18: Petrol and diesel excise duty rates (€2023/litre) and implied carbon tax (€2023/tCO2)

Country	Petrol Duty	Diesel Duty	Implied Carbon Tax Petrol	Implied Carbon Tax Diesel
Austria	0.48	0.40	207.79	149.25
Belgium	0.60	0.60	259.74	223.88
Bulgaria	0.36	0.33	155.84	123.13
Croatia	0.41	0.35	177.49	130.60
Cyprus	0.43	0.40	186.15	149.25
Denmark	0.64	0.44	277.06	164.18
Estonia	0.56	0.37	242.42	138.06
EU Minimum	0.36	0.33	155.41	123.13
Finland	0.72	0.53	311.69	197.76
France	0.68	0.59	294.37	220.15
Germany	0.67	0.49	290.04	182.84
Greece	0.70	0.41	303.03	152.99
Hungary	0.29	0.28	125.54	104.48
Ireland	0.48	0.43	207.79	160.45
Italy	0.73	0.62	316.02	231.34
Latvia	0.51	0.41	220.78	152.99
Lithuania	0.47	0.37	203.46	138.06
Luxembourg	0.54	0.41	233.77	152.99
Malta	0.36	0.33	155.84	123.13
Netherlands	0.79	0.52	341.99	194.03
Norway	0.50	0.33	218.18	122.39
Poland	0.37	0.33	160.17	123.13
Portugal	0.55	0.42	238.10	156.72
Romania	0.36	0.33	155.84	123.13
Slovakia	0.51	0.37	220.78	138.06
Slovenia	0.47	0.51	203.46	190.30
Spain	0.50	0.38	216.45	141.79
Sweden	0.58	0.38	251.08	141.79
Switzerland	0.98	1.01	424.24	375.00
UK	0.62	0.62	268.40	231.34

Notes: To estimate the implied carbon tax associated with fuel duties, standard carbon dioxide (CO_2) emission factors for petrol and diesel were sourced from the Intergovernmental Panel on Climate Change (IPCC) 2006 Guidelines for National Greenhouse Gas Inventories (IPCC, 2006). The emission intensities used were: Petrol (gasoline): 2.31 kg CO_2 per litre; Diesel: 2.68 kg CO_2 per litre

Lastly, in the carbon price scenarios, subsidies for key renewable technologies at the MS level (Table A. 19) are modelled, and it is assumed that these subsidies are gradually phased out by 2050.

Table A. 19: Assumed subsidies (€'23/MWh-e except for bioenergy, which is in €'23/MWh-th) for RES technologies in the Baseline, High and Low carbon price scenarios

	FiT tariffs			
	Solar	Wind	Bioenergy	Hydro
Baltics	34.49	7.95	3.80	-
BE	90.87	36.44	4.97	18.73
Central Europe	72.06	-	24.26	0.38
DE	71.93	30.57	17.64	6.13
Eastern Europe	111.66	-	3.54	0.63
FR	87.16	19.39	24.28	2.19
Iberia	49.09	0.19	2.94	-
Ireland	88.93	13.67	28.03	-
IT	108.96	29.56	21.98	9.09
NL	53.66	29.08	-	-
Nordic	46.51	0.31	4.55	-
PL	37.30	-	0.29	-
Southeast Europe (SEE)	27.62	-	3.29	-

Source: own calculations based on Enerdata, Trinomics, 2023[28]
Notes: These tariffs are calibrated such that the resulting total aligns with the 2021 total subsidies for the EU

[28] https://op.europa.eu/en/publication-detail/-/publication/32d284d1-747f-11ee-99ba-01aa75ed71a1/language-en

Appendix 4 – Model Calibration Results

This Appendix first presents the calibration results for 2030 and 2050. It compares these results with the EC's 2030 MIX and 2050 NZ (1.5 TECH) at the European level and for key MS. The objective of presenting these brief comparative analyses is to show the alignment and differences of the modelling results presented with the official modelling presented by the EC.

The analysis of power generation capacity (Figure A. 3, upper left panel for 2030 and lower left panel for 2050) and final energy consumption (Figure A. 3, upper right panel for 2030 and lower right panel for 2050) highlights key areas of alignment and differences between the model results and the European Commission's MIX2030 and 1.5 TECH 2050 scenarios. One of the most critical factors influencing these differences is the structural variation between the model results and the scenarios: the model results encompass the EU27, UK, Norway, and Switzerland, while the MIX2030 scenario covers only the EU27 and the EC 1.5 TECH scenario includes EU27 and the UK, which have distinct energy policies and resource endowments (see discussion in Appendix A.2.9). This is likely to contribute to the higher overall capacities and differences in specific energy sources. For example, the total installed capacity in 2030 is 9.0% higher in the model results (1,456 MW) compared to MIX2030 (1,336 MW), with specific increases observed in natural gas (+22.6%) and wind offshore (+42.4%). This structural difference partially explains the observed discrepancies, particularly in the installed capacities of renewables and fossil fuels.

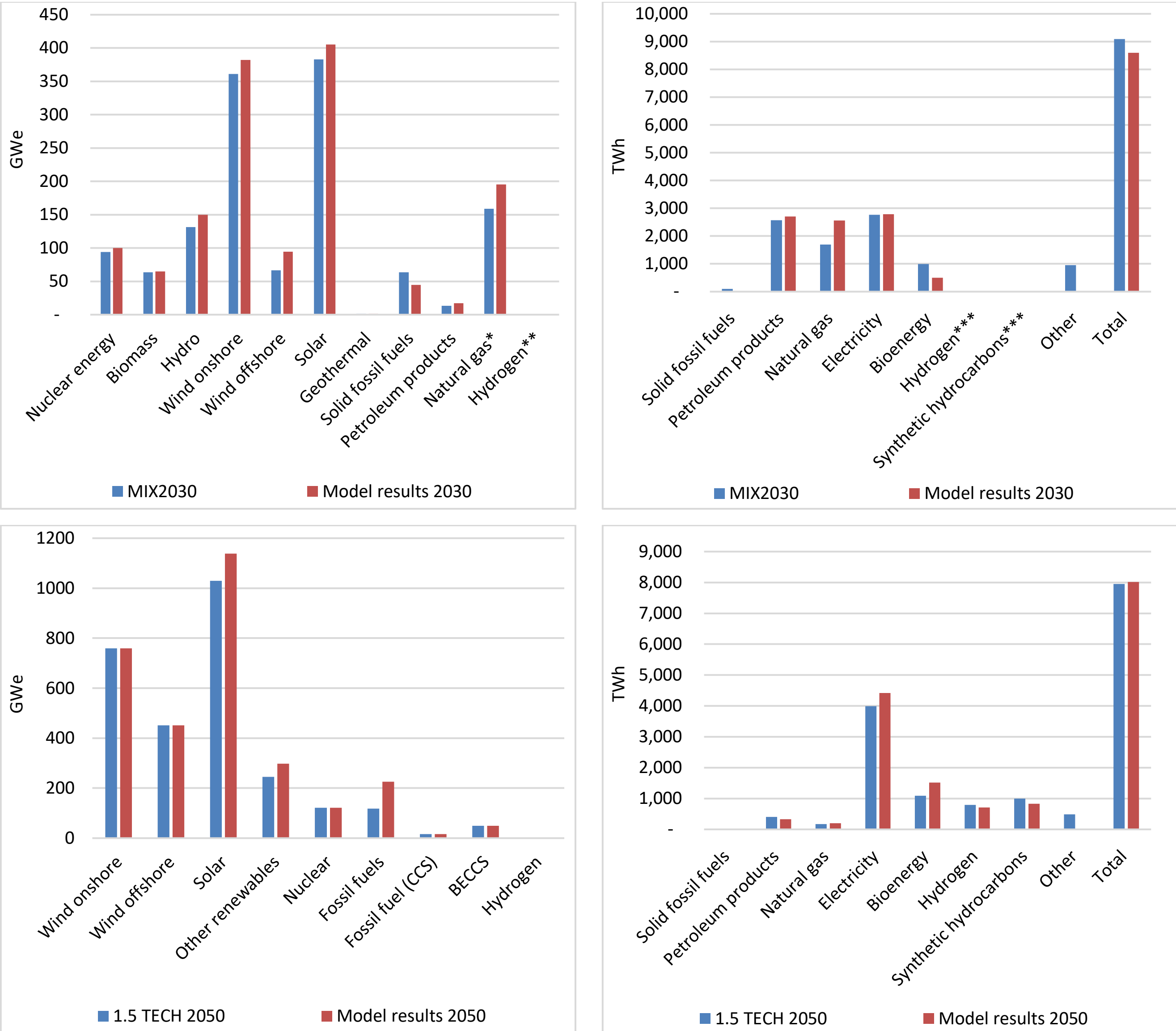
Methodological differences also play a role in the observed variations. The model employed in this study is an hourly model with global gas trade endogenously modelled, allowing for a more nuanced capture of the role of gas in energy systems with an increasing share of intermittent renewable sources. This model feature contrasts with the EC scenarios, which may not fully account for such detailed dynamics. For instance, the model results for 2030 show a significantly higher share of natural gas in final energy consumption (+50.6% compared to MIX2030), reflecting its role in complementing the higher share of variable renewables like solar and wind. The model anticipates that natural gas will provide critical backup capacity, especially as renewables become a more significant part of the energy mix, which may not be as explicitly captured in the MIX2030 scenario.

In 2050, both the model results and the 1.5 TECH scenario indicate a significant shift towards renewable energy, with solar and wind contributing to a majority of the power generation capacity (solar: 37.2% in the model results vs. 36.9% in 1.5 TECH; wind onshore and offshore combined: 39.5% in the model results vs. 43.4% in 1.5 TECH). However, the model shows a substantial difference in fossil fuel (unabated natural gas) capacity without carbon capture and storage (CCS), which is 91.5% higher than in the 1.5 TECH scenario (226 MW vs. 118 MW). This suggests that the model assumes a continued, albeit reduced, role for natural gas capacity to ensure energy security, possibly due to the inclusion of endogenous global gas trade, hourly modelling of demand and renewable supply, and broader geographical scope. Additionally, the model projects a 38.9% higher share of bioenergy in final consumption (1,518 TWh vs. 1,093 TWh in 1.5 TECH), indicating a greater reliance on bioenergy, possibly due to more favourable assumptions about biomass availability or lower-cost resources than what EC scenarios assumed.

When comparing final energy consumption between 2030 and 2050, the model results suggest a consistent trajectory towards net zero GHG emissions. However, the differences in the shares of various fuels highlight the structural differences in the modelling approach and input assumptions. For instance, the model results for 2030 show a significant reduction in solid fossil fuel consumption (-84.2% compared to MIX2030), while natural gas consumption increases (+50.6% compared to MIX2030). By 2050, the model continues to emphasise electrification, with electricity's share in final consumption rising to 55.2% (4,420 TWh), compared to 50.1% (3,989 TWh) in 1.5 TECH. This increase reflects the model's assumption that electricity will be the primary energy carrier in a decarbonised economy driven by widespread electrification across sectors.

The analysis underscores the importance of considering structural and methodological differences when interpreting the results. In particular, the modelling results are dependent on three sets of factors: (i) carbon budget constraints for the energy system as a whole as well as sector-specific constraints (i.e., buildings and road transport sectors; see Appendix A2.7), carbon price and renewable subsidies (see Appendix 3), (ii) technology costs and techno-economic characteristics (see Appendix A.2.3), and (iii) resource availability assumptions (see Appendix A.2.8 and A.2.9) and their costs. The inclusion of additional countries in the model and the more detailed treatment of global gas trade dynamics contribute to the observed differences, particularly in the roles of natural gas and renewables. The substantial differences in fossil fuel (unabated natural gas) capacity and bioenergy consumption between the model results and the EC scenarios also reflect varying assumptions about the pace of technological deployment, energy security needs, and the availability of resources at the MS level.

Figure A. 3: Generation mix (left) and final consumption (right) in 2030 (first row) and in 2050 (second row) at the European level



Source: MIX2030 taken from https://energy.ec.europa.eu/data-and-analysis/energy-modelling/policy-scenarios-delivering-european-green-deal_en; 1.5 TECH taken from https://climate.ec.europa.eu/system/files/2019-08/long-term_analysis_in_depth_analysis_figures_20190722_en.pdf

Notes: * Natural and manufactured gases; **Hydrogen and synthetic hydrocarbons; ***Low-carbon; model results cover EU27, UK, NO and CH Note that the generation mix and final consumption in MIX 2030 scenario cover only EU27 only; hence, they are strictly not comparable to those from the EC MIX 2030 scenario. Note that the generation mix and final consumption in 1.5 TECH covers only EU27 and UK, so the modelling results are not strictly comparable to those from the EC 1.5 TECH scenario

The selected country-level results show similar patterns of differences between the scenarios. The EC's 2050 NZ scenario did not publish results at the Member State level, so a comparison between our modelling and official results is only focused on 2030 (see Figure A. 4).

The analysis of the differences between the 2030 power generation mix in the model results and the European Commission's MIX2030 scenario highlights areas of both alignment and divergence across different technologies and countries. These differences are driven by variations in resource strengths, Feed-in Tariffs (FiTs), and resource bounds that are considered in this model, as well as structural and methodological differences in the modelling approaches.

Regarding similarities, the model results align closely with MIX2030 in deploying nuclear energy and natural gas across key countries. For example, nuclear energy capacities in France and natural gas in Germany and Italy are within the +/-10% threshold compared to MIX2030, reflecting perhaps aligned assumptions about these technologies. This alignment suggests that the model results closely match the European Commission's outlook in areas where resource strengths and policy frameworks are consistent.

However, significant differences are evident in deploying renewable energy technologies, particularly in countries like Germany and the Netherlands. The model results show a 90% increase in onshore wind capacity in Germany and a 92.3% increase in offshore wind capacity in the Netherlands compared to MIX2030. These differences are influenced by higher-than-average resource strengths and FiTs in these countries. For instance, Germany's onshore wind resource strength is 5.8% above the average, with a FiT 60% higher than the average, making onshore wind a highly favourable option. In the Netherlands, the offshore wind resource strength is 13.6% above the average, and the FiT is 52% higher, justifying the model's emphasis on maximising wind energy in these regions.

In contrast, the model underutilises onshore wind in France and Poland, where capacities are reduced by 4.9% and 55.6%, respectively, compared to MIX2030. In France, the resource strength for onshore wind is 15.6% below the average, and the FiT is only marginally higher, making this technology less competitive. In Poland, the absence of a FiT in the model (due to input data based on a report by Enerdata and Trinomics, 2023[29]) negates the advantage of a resource strength that is 7.4% above the average, leading to a significant reallocation of wind capacity to other regions with more supportive policies.

The model also shows a notable increase in solar energy capacity in Italy, with a 23.2% rise compared to MIX2030. This is due to Italy's solar resource strength being 7.7% above the average and an exceptionally high FiT, 83.3% above the average, making solar more attractive than other renewables in the region.

Methodologically, one key difference influencing these results is the detailed hourly modelling approach used in the current analysis, which captures the variability of renewable energy sources and the role of global gas trade more accurately. This approach allows the model to better account for the intermittency of renewables and the need for flexible backup capacity, leading to a more dynamic allocation of resources compared to the more static assumptions in the MIX2030 scenario. Additionally, the current model's ability to endogenously model global gas trade provides a more nuanced understanding of the natural gas role in complementing variable renewables, which may not be fully captured in the MIX2030 scenario.

In conclusion, while significant areas of alignment exist between the model results and MIX2030, particularly for established technologies like nuclear and natural gas, notable differences arise in deploying renewable energy technologies. These differences are driven by variations in the assumed renewables resource strengths (capacity factors), FiTs, and the model's methodological approach. The dynamic and detailed nature of the current model, which includes hourly dispatch and global gas trade considerations, results in a more flexible and responsive allocation of resources, leading to deviations from MIX2030, especially where resource strengths and financial incentives vary significantly from the average.

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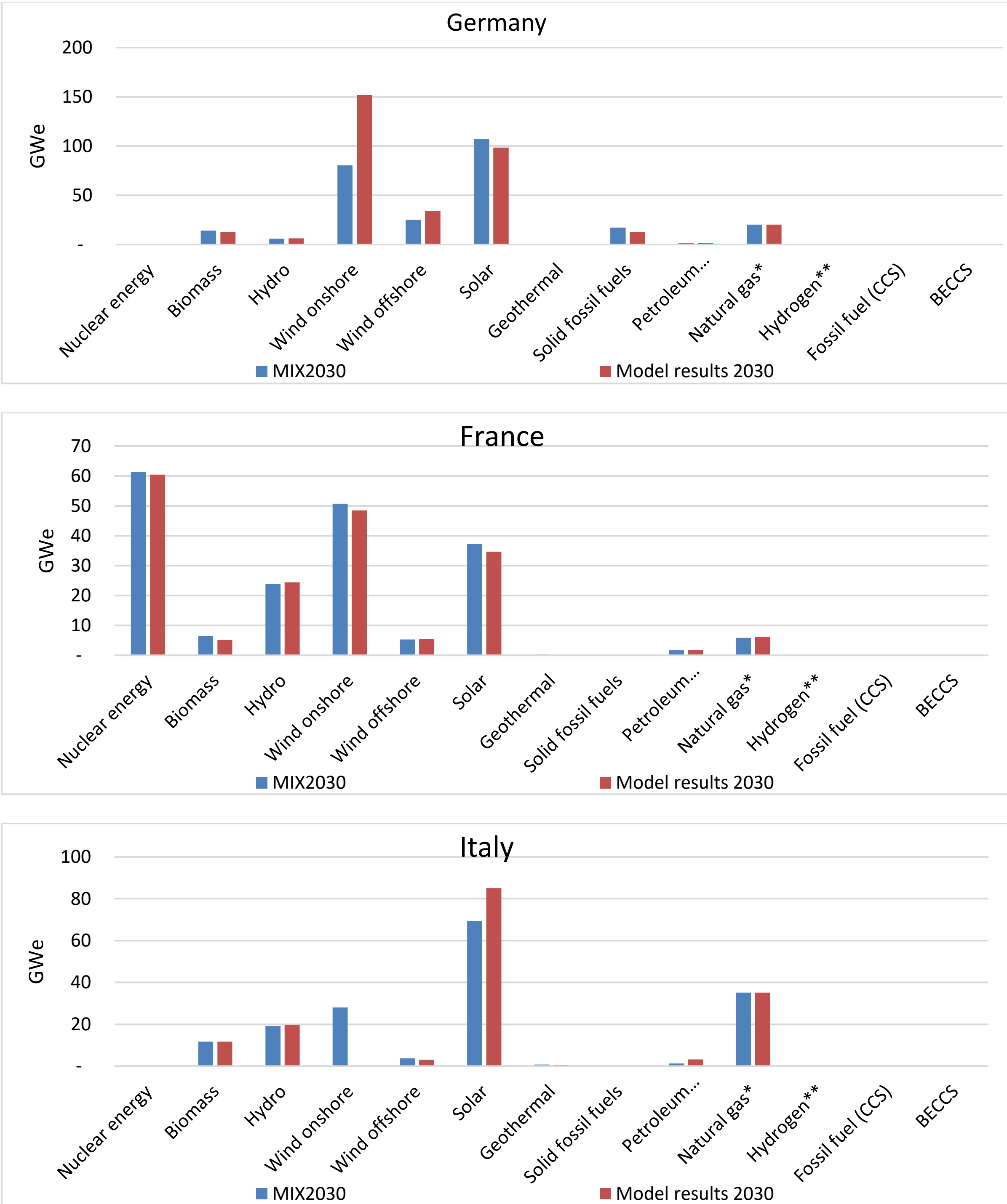
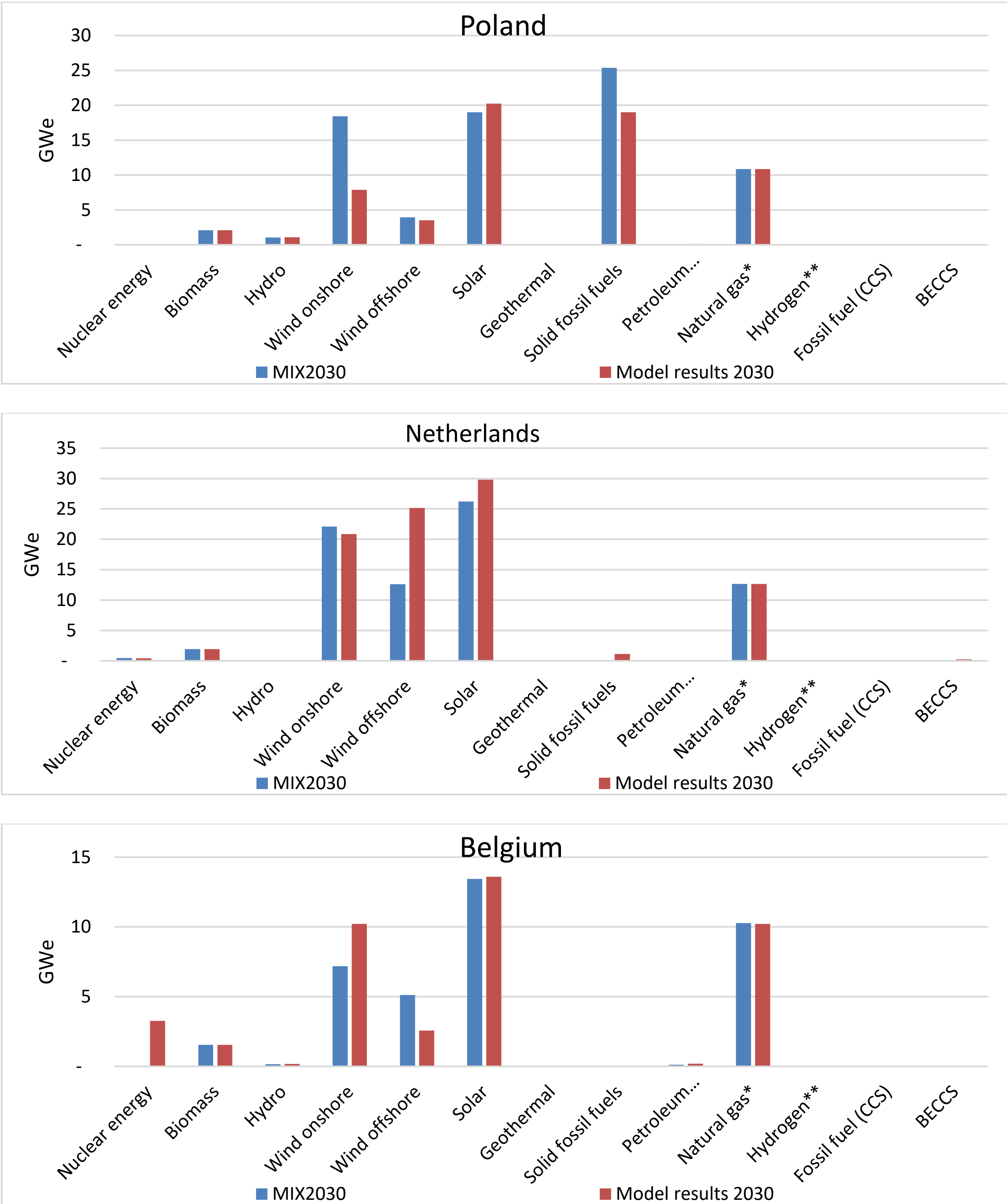
[29] https://op.europa.eu/en/publication-detail/-/publication/32d284d1-747f-11ee-99ba-01aa75ed71a1/language-en

Figure A. 4: Generation mix (GWe) of selected countries in 2030





Source: MIX2030 taken from https://energy.ec.europa.eu/data-and-analysis/energy-modelling/policy-scenarios-delivering-european-green-deal_en
Notes: * Natural and manufactured gases; **Hydrogen and synthetic hydrocarbons. Note that in 2024